# Block practice · Diet, digestion and teeth

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Match digestive enzymes to substrate, product and site

Identify alimentary canal organs and describe what each does

Link a nutrient to its function, source and deficiency disease

Name tooth types and tissues, state their jobs and compare tooth sets

Define a balanced diet and explain why needs differ between people

Explain how tooth decay happens and justify ways to prevent it

Explain an enzyme practical result, including levelling off as substrate runs out

Explain how villi aid absorption and the effect of damage

Explain where bile comes from and how it aids fat digestion

Measure food energy by burning and explain why the value is too low

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# Practice workbook · Match digestive enzymes to substrate, product and site

Amylase, maltase, proteases and lipase: their substrates, products and where they are made or act.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DIGESTIVE-ENZYME-MATCH

# 1. Question 2 · part a.i · 3 marks

June 2023 Paper 2R

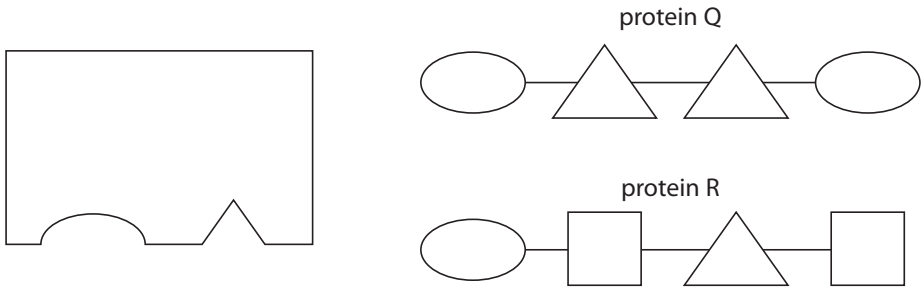
Worked-example retry · Attempt part a.i only.

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**2** (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.




**Enzyme P** **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

**(3)**

| | | | | |
|---|---|---|---|---|
| amino acids | amylases | fatty acids | proteases | |
| protein Q | protein R | starch | sugar | |

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

**(1)**

.....

(iii) Name the molecule that codes for the production of enzyme P.

**(1)**

.....

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## 1. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

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## 2. Question 3 · part b · 1 mark

June 2019 Paper 2

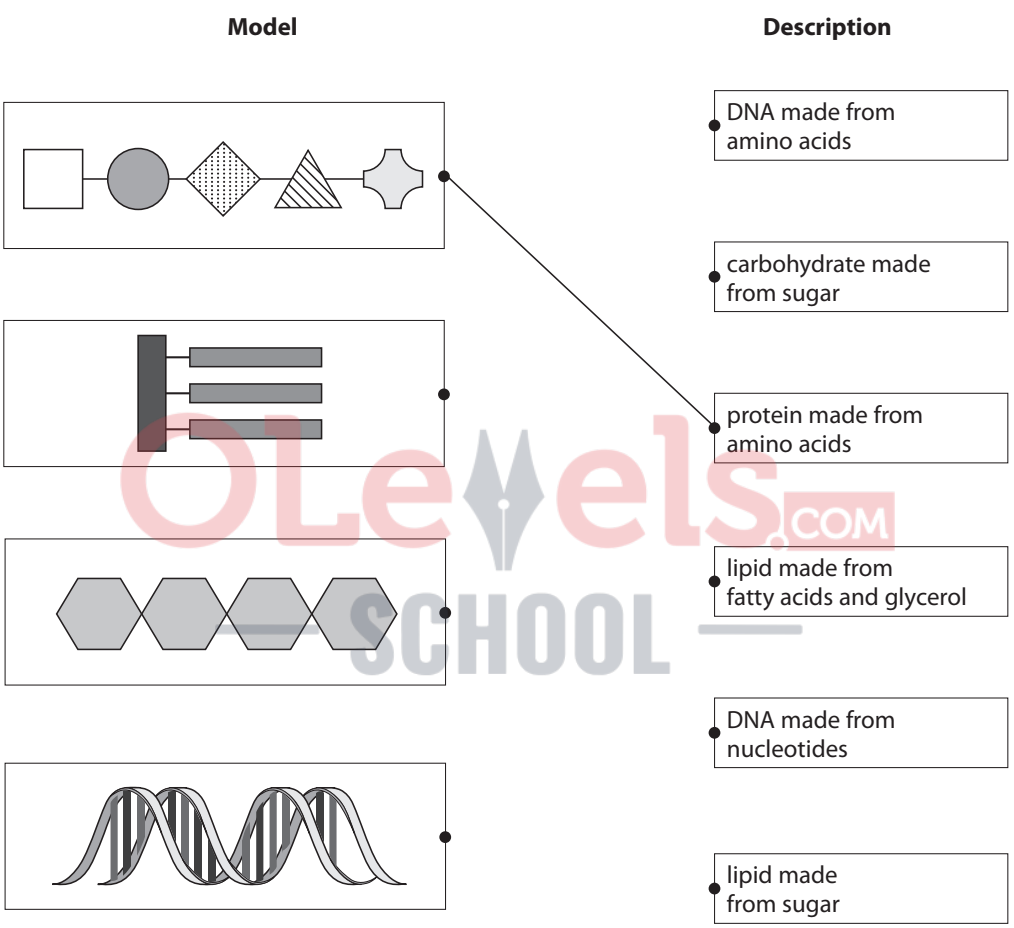
Further practice · Attempt part b only.

**3** (a) The diagram shows models of four biological molecules.

Draw a straight line from each model to its correct description.

One has been done for you.

**(3)**




(b) Which body organ produces enzymes that break down protein?

**(1)**

- ☒ **A** gall bladder
- ☒ **B** large intestine
- ☒ **C** mouth
- ☒ **D** stomach

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P 5 8 5 6 6 A 0 6 2 8

## 2. Question 3 · part b · 1 mark

June 2019 Paper 2

Further practice · Attempt part b only.

(c) A student uses this method to test different substances for protein.

- grind solid substances into small pieces
- place substances into separate test tubes
- add a few drops of Biuret reagent to each substance
- record the colour change for each substance

The table shows the student's results.

| Substance | Colour after Biuret test |
|---|---|
| milk | purple |
| pasta | blue |
| lemon juice | blue |
| cheese | purple |
| distilled water | purple |

(i) In the student's test, how many of the substances give a positive result for protein?

**(1)**

.....

(ii) The result for distilled water is incorrect.

Suggest one mistake the student could have made to get this incorrect result.

**(1)**

.....

.....

(d) The table shows an incomplete risk assessment for the Biuret test.

Complete the table by describing how to reduce the risk of each hazard.

**(2)**

| Hazard | Reducing risk |
|---|---|
| stools – trip hazard | keep stools under bench |
| broken glass – cuts | |
| Biuret reagent – irritant | |

**(Total for Question 3 = 8 marks)**

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## 2. Question 3 · part b · 1 mark · Working and answer

Working and answer

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## 3. Question 1 · part b.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.i only.

**1** Food enters the alimentary canal through the mouth.

All undigested food leaves the body through the anus.

(a) Name three structures the food passes through between the mouth and the anus.

**(3)**

1 .....

2 .....

3 .....

(b) Some of the food is digested by enzymes as it passes through the alimentary canal.

(i) Name an enzyme that digests protein.

**(1)**

.....

(ii) Describe the processes that occur in the mouth to digest food.

**(4)**

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2

P 6 2 0 5 4 A 0 2 2 4

## 3. Question 1 · part b.i · 1 mark · Working and answer

Working and answer

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# 4. Question 2 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

**2** A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.

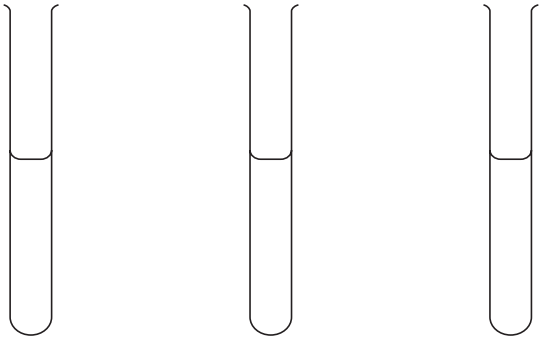



Table 1 lists the contents of each test tube.

| Test tube A | Test tube B | Test tube C |
|---|---|---|
| 5 $cm^3$ of milk | 5 $cm^3$ of milk | 5 $cm^3$ of milk |
| 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate |
| 6 drops of phenolphthalein | 6 drops of phenolphthalein | 6 drops of phenolphthalein |
| 1 $cm^3$ enzyme | 1 $cm^3$ enzyme | 1 $cm^3$ boiled enzyme |
| distilled water | bile salts | bile salts |

**Table 1**

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# 4. Question 2 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

| | Colour of contents | | |
|---|---|---|---|
| | **Tube A** | **Tube B** | **Tube C** |
| at start | pink | pink | pink |
| after 5 minutes | pink | colourless | pink |
| after 10 minutes | pink | colourless | pink |
| after 15 minutes | colourless | colourless | pink |

**Table 2**

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation?

**(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids?

**(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced?

**(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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## 4. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

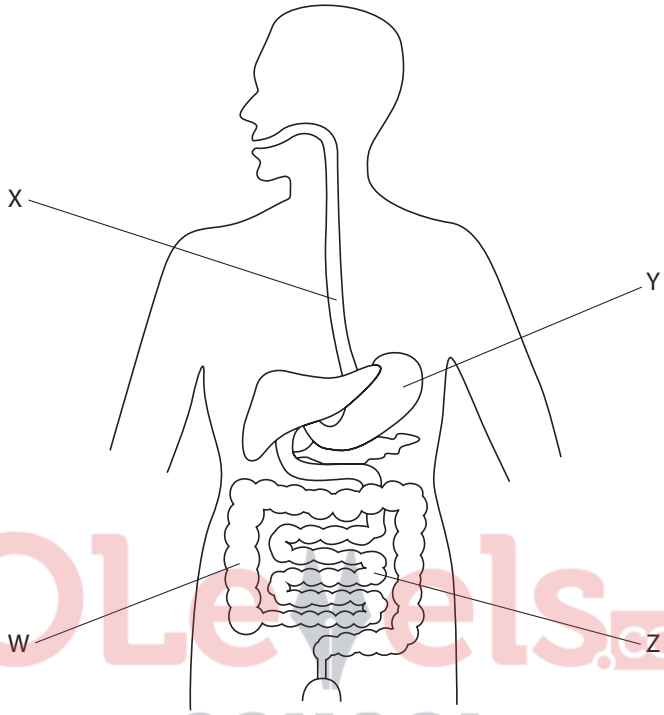
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# 5. Question 2 · part b.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.i only.

**2** (a) The diagram shows the human digestive system.




(i) Name the structures labelled W, X, Y and Z.

**(4)**

W .....

X .....

Y .....

Z .....

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P 7 0 9 4 9 A 0 6 2 4

# 5. Question 2 · part b.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.i only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

| | | | |
|---|---|---|---|
| relax | coordination | contract | liquid |
| peristalsis | muscles | enzymes | bolus |

Use words from the box to complete the passage.

**(4)**

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

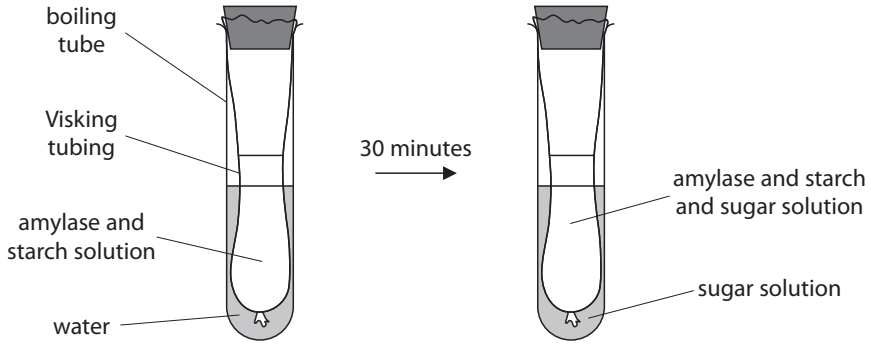
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# 5. Question 2 · part b.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.i only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.




(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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## 5. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

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# 6. Question 4 · part a.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a.ii only.

**4** (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

**(3)**

(ii) Name the enzyme that digests lipids.

**(1)**

(iii) Name the products of lipid digestion.

**(2)**

(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

**Temperature at 37 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 8.5 | 8.0 | 7.5 | 7.2 | 7.0 | 6.8 | 6.2 | 5.8 | 5.5 | 5.5 |

**Temperature at 80 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 9.2 | 9.0 | 8.8 | 8.8 | | 8.6 | 8.6 | 8.4 | 8.4 | 8.4 |

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## 6. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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## 7. Question 4 · part a.iii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part a.iii only.

**4** (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

**(3)**

(ii) Name the enzyme that digests lipids.

**(1)**

(iii) Name the products of lipid digestion.

**(2)**

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(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

**Temperature at 37 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 8.5 | 8.0 | 7.5 | 7.2 | 7.0 | 6.8 | 6.2 | 5.8 | 5.5 | 5.5 |

**Temperature at 80 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 9.2 | 9.0 | 8.8 | 8.8 | | 8.6 | 8.6 | 8.4 | 8.4 | 8.4 |

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## 7. Question 4 · part a.iii · 2 marks · Working and answer

Working and answer

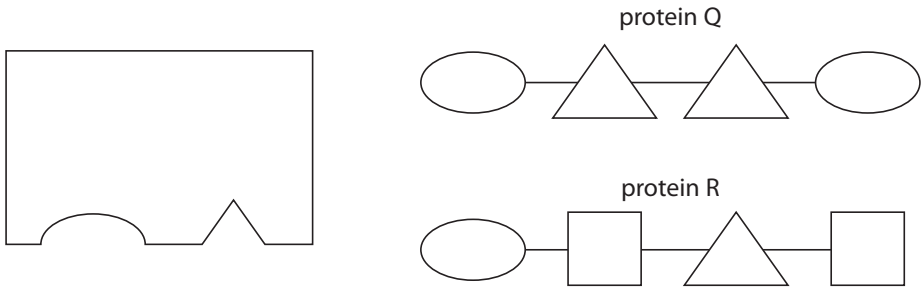
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# 8. Question 2 · part a.ii · 1 mark

June 2023 Paper 2R

Further practice · Attempt part a.ii only.

**2** (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.




**Enzyme P** **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

**(3)**

| | | | |
|---|---|---|---|
| amino acids | amylases | fatty acids | proteases |
| protein Q | protein R | starch | sugar |

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

**(1)**

.....

(iii) Name the molecule that codes for the production of enzyme P.

**(1)**

.....

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## 8. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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# 9. Question 4 · part a.vi · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.vi only.

**4** (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

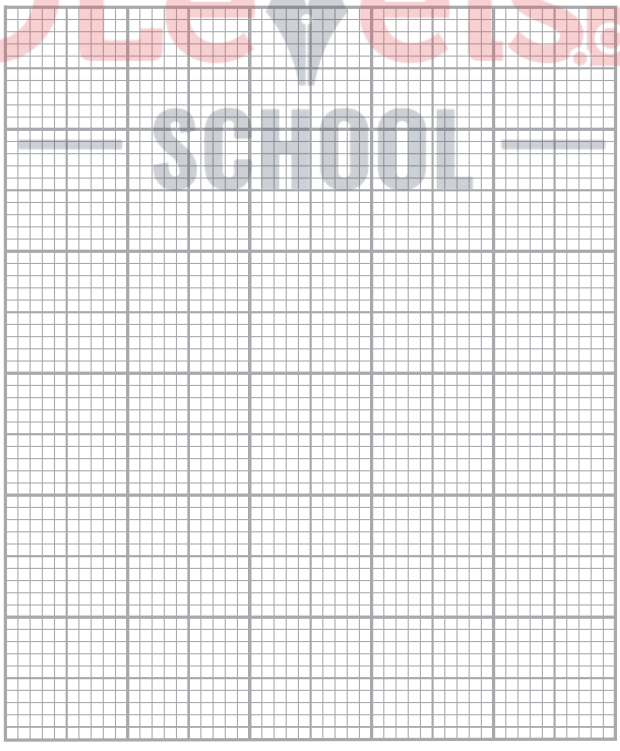
| pH | Time taken to digest starch in minutes |
|---|---|
| 2 | 60 |
| 4 | 45 |
| 6 | 20 |
| 8 | 35 |
| 10 | 55 |

(i) Plot a graph using the data in the table.

**(3)**

(ii) Draw a curve of best fit.

**(1)**



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## 9. Question 4 · part a.vi · 1 mark

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Further practice · Attempt part a.vi only.

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(iii) Use the graph to determine the optimum pH for the digestion of starch.

**(1)**

(iv) Describe the trend shown by the graph.

**(2)**

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

**(3)**

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(vi) Name a part of the body where starch digestion takes place.

**(1)**

(b) Describe a test for starch in a sample of food.

**(2)**

**(Total for Question 4 = 13 marks)**

## 9. Question 4 · part a.vi · 1 mark · Working and answer

Working and answer

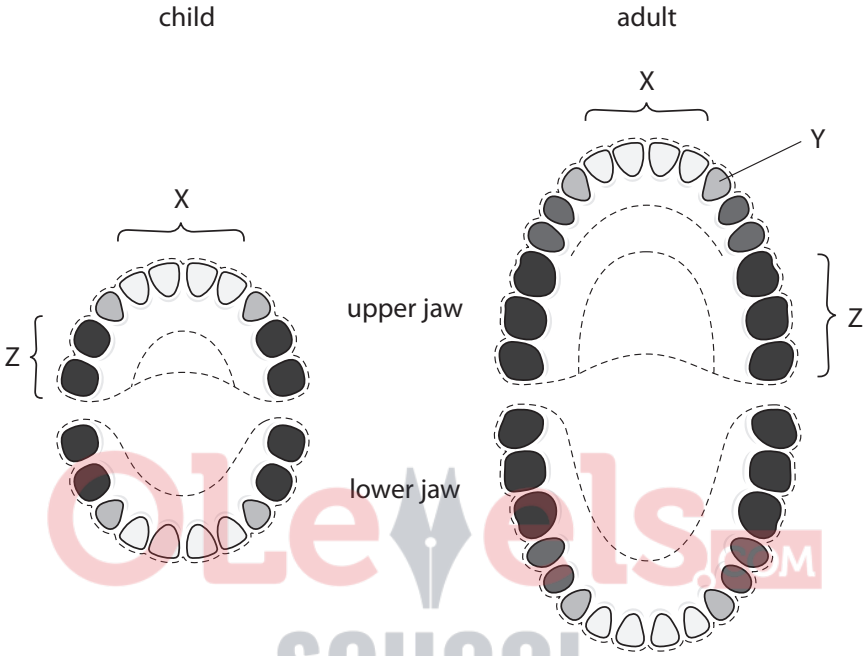
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# 10. Question 1 · part b · 4 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

**1** (a) The diagrams show the arrangement of teeth in a young child and the arrangement of teeth in an adult.




(Source: https://www.medicalnewstoday.com/articles/326754#number-of-teeth)

(i) Name the types of teeth shown by X and Z.

**(2)**

X .....

Z .....

(ii) State the function of the tooth labelled Y.

**(1)**

.....

.....

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P 7 4 4 9 5 A 0 2 2 8

## 10. Question 1 · part b · 4 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

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(b) Digestion of food begins in the mouth.

Use the most appropriate words or numbers to complete the passage about digestion of food in the mouth.

**(4)**

Saliva in the mouth contains an enzyme called ..... .

This enzyme speeds up the digestion of starch to produce

..... .

The enzyme in saliva works best at pH ..... .

If the pH in the mouth was too low or too high, the digestion of carbohydrates

would ..... .

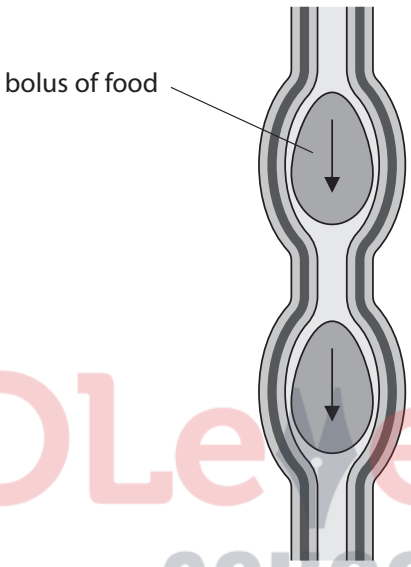
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## 10. Question 1 · part b · 4 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

(c) The diagram shows the part of the digestive system that transports food from the mouth to the stomach.




(Source: https://www.motifolio.com/1031265.html)

(i) Name the part of the digestive system that transports food from the mouth to the stomach.

**(1)**

.....

(ii) Describe how the action of this part of the digestive system transports food from the mouth to the stomach.

**(2)**

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**(Total for Question 1 = 10 marks)**

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## 10. Question 1 · part b · 4 marks · Working and answer

Working and answer

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## 11. Question 7 · part f · 2 marks

November 2024 Paper 2

Further practice · Attempt part f only.

**7** Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

| Student | Volume of enzyme solution in $cm^3$ | Temperature in °C | pH | Volume of cloudy suspension in $cm^3$ | Time tubes left in minutes | Appearance at end of investigation |
|---|---|---|---|---|---|---|
| 1 | 5 | 37 | 2 | 10 | 9 | cloudy |
| 2 | 5 | 37 | 2 | 10 | 20 | clear |
| 3 | 5 | 37 | 4 | 10 | 35 | clear |
| 4 | 5 | 37 | 4 | 10 | 20 | cloudy |
| 5 | 5 | 37 | 6 | 10 | 8 | cloudy |
| 6 | 5 | 37 | 6 | 10 | 40 | cloudy |

(a) Explain why the investigation was carried out at 37 °C.

**(2)**

(b) Explain which property of the enzyme the students are investigating.

**(2)**

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## 11. Question 7 · part f · 2 marks

November 2024 Paper 2

Further practice · Attempt part f only.

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(c) Comment on the results of the investigation.

In your answer include ideas of validity and accuracy.

**(4)**

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(d) Describe a control experiment that the students could have carried out.

**(3)**

## 11. Question 7 · part f · 2 marks

November 2024 Paper 2

Further practice · Attempt part f only.

(e) State a possible conclusion that the students could make from the results of this investigation.

**(2)**

(f) Explain where this enzyme is likely to be found in the alimentary canal.

**(2)**

**(Total for Question 7 = 15 marks)**

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Working and answer

## 12. Question 7 · part a · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a only.

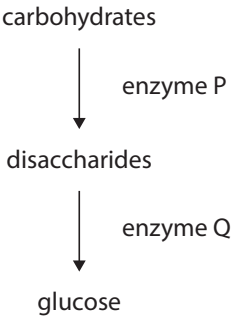
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**7** (a) The diagram shows how carbohydrates are broken down to glucose by two different enzymes, enzyme P and enzyme Q.

These enzymes are found in different regions of the digestive system.




Name enzyme P.

**(1)**

.....

(b) People with diabetes can use glucose testing strips to test their urine.

Describe how enzymes are used in glucose testing strips.

**(2)**

.....

.....

.....

.....

(c) People with type 1 diabetes are unable to produce hormones to control their blood glucose levels.

Name the organ responsible for producing hormones to control levels of glucose in the blood.

**(1)**

.....

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P 7 4 6 3 2 A 0 1 9 2 4

## 12. Question 7 · part a · 1 mark · Working and answer

Working and answer

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# Practice workbook · Identify alimentary canal organs and describe what each does

Naming, labelling or locating digestive organs from a diagram or feature, and describing the job of the mouth, pancreas, small or large intestine.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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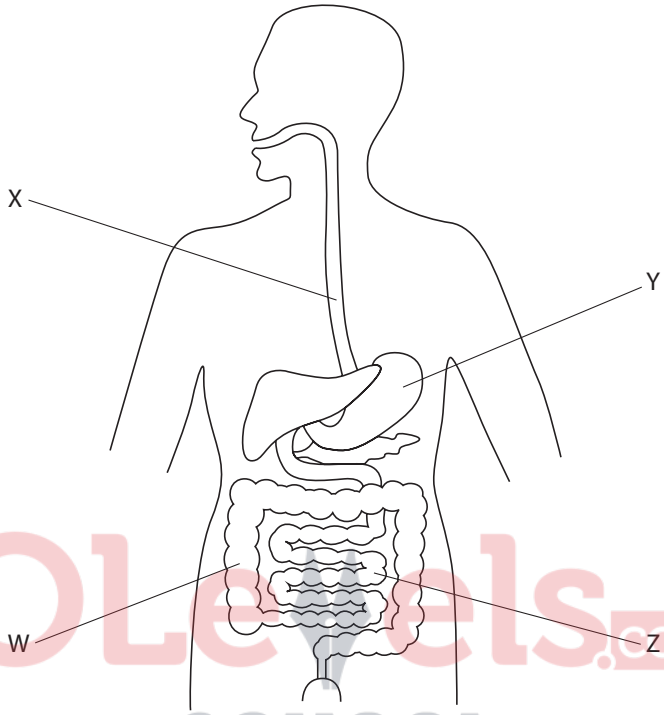
Type ID: HBI-GUT-ORGANS-IDENTIFY-AND-ROLE

# 1. Question 2 · part a.i · 4 marks

June 2022 Paper 1R

Worked-example retry · Attempt part a.i only.

**2** (a) The diagram shows the human digestive system.




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(i) Name the structures labelled W, X, Y and Z.

**(4)**

W .....

X .....

Y .....

Z .....

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# 1. Question 2 · part a.i · 4 marks

June 2022 Paper 1R

Worked-example retry · Attempt part a.i only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

| | | | |
|---|---|---|---|
| relax | coordination | contract | liquid |
| peristalsis | muscles | enzymes | bolus |

Use words from the box to complete the passage.

**(4)**

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

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## 1. Question 2 · part a.i · 4 marks · Working and answer

Working and answer

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## 2. Question 1 · part whole · 6 marks

November 2020 Paper 1

Further practice · Attempt part whole only.

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**1** The table lists some features and regions of the human alimentary canal.

Tick (✓) one box in each row to show the region of the alimentary canal where each feature occurs.

**(6)**

| Feature | Region of alimentary canal | | | |
|---|---|---|---|---|
| | **mouth** | **stomach** | **small intestine** | **large intestine** |
| starts protein digestion | | | | |
| starts carbohydrate digestion | | | | |
| has a pH of 2 | | | | |
| has villi | | | | |
| absorbs most water | | | | |
| connects with bile duct | | | | |

**(Total for Question 1 = 6 marks)**

OLevels.com SCHOOL

## 2. Question 1 · part whole · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 3. Question 1 · part a · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a only.

**1** Food enters the alimentary canal through the mouth.

All undigested food leaves the body through the anus.

(a) Name three structures the food passes through between the mouth and the anus.

**(3)**

1 .....

2 .....

3 .....

(b) Some of the food is digested by enzymes as it passes through the alimentary canal.

(i) Name an enzyme that digests protein.

**(1)**

.....

(ii) Describe the processes that occur in the mouth to digest food.

**(4)**

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OLevels.com SCHOOL

## 3. Question 1 · part a · 3 marks · Working and answer

Working and answer

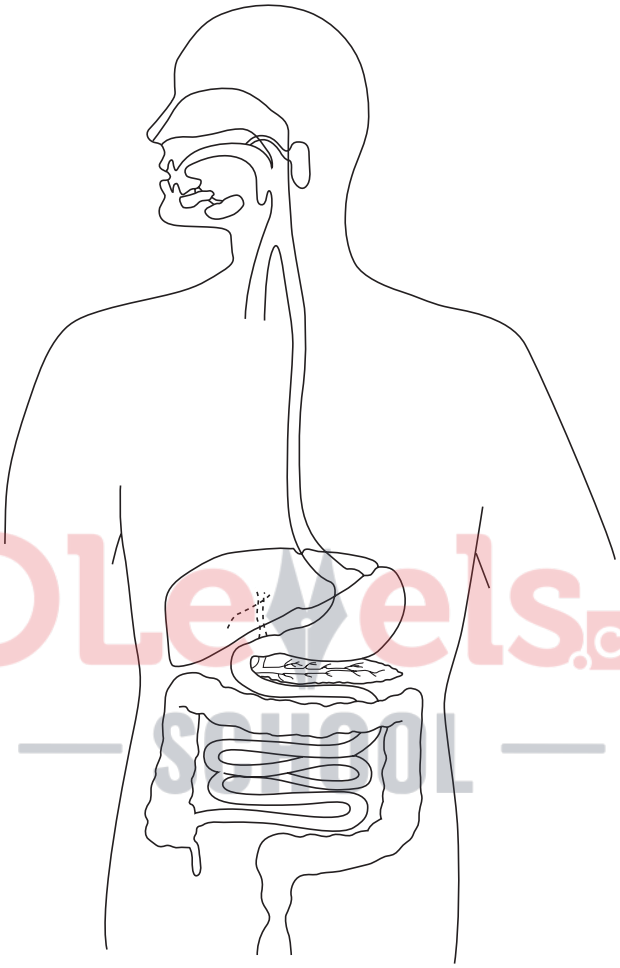
OLevels.com SCHOOL

# 4. Question 4 · part a.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.i only.

**4** The diagram shows the alimentary canal.



(a) Identify these areas of the alimentary canal using label lines and the correct letters.

(i) Label with an X the area where starch digestion begins.

**(1)**

(ii) Label with a Y the area that has a pH of 2

**(1)**

(iii) Label with a Z the area where both lipid and protein digesting enzymes are produced.

**(1)**

DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA



P 6 7 0 6 0 R R A 0 8 2 0

## 4. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

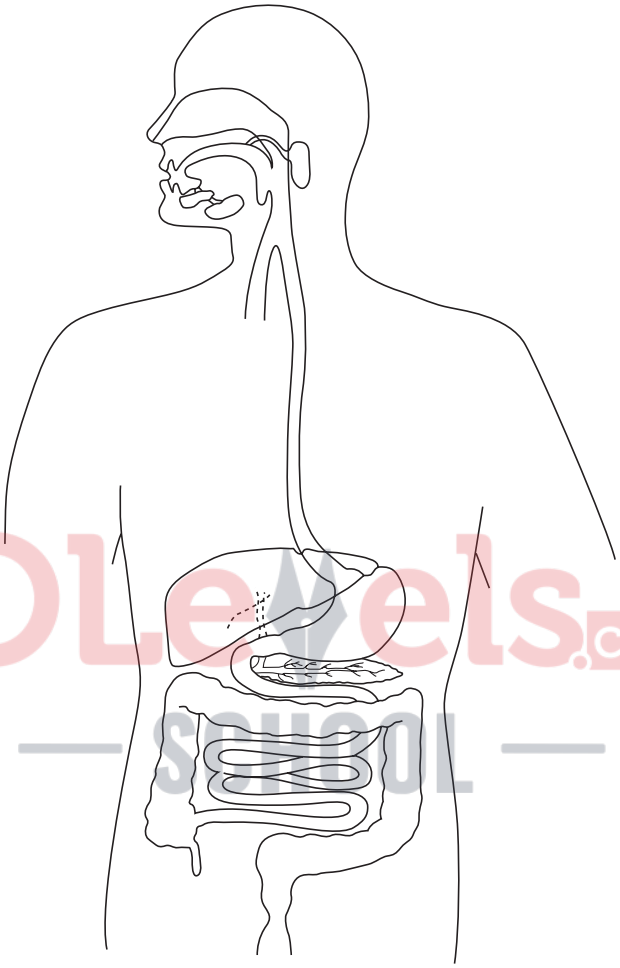
OLevels.com SCHOOL

## 5. Question 4 · part a.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.ii only.

**4** The diagram shows the alimentary canal.



(a) Identify these areas of the alimentary canal using label lines and the correct letters.

(i) Label with an X the area where starch digestion begins.

**(1)**

(ii) Label with a Y the area that has a pH of 2

**(1)**

(iii) Label with a Z the area where both lipid and protein digesting enzymes are produced.

**(1)**

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## 5. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

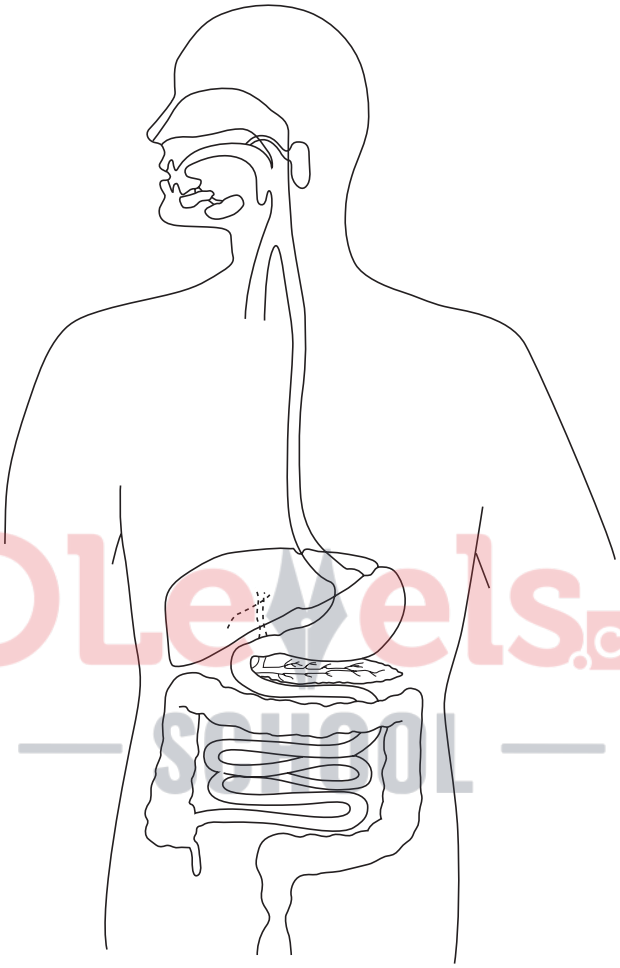
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## 6. Question 4 · part a.iii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.iii only.

**4** The diagram shows the alimentary canal.



(a) Identify these areas of the alimentary canal using label lines and the correct letters.

(i) Label with an X the area where starch digestion begins.

**(1)**

(ii) Label with a Y the area that has a pH of 2

**(1)**

(iii) Label with a Z the area where both lipid and protein digesting enzymes are produced.

**(1)**

DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA



P 6 7 0 6 0 R R A 0 8 2 0

## 6. Question 4 · part a.iii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

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THIS AREA

(b) Some people have the condition known as coeliac disease. This results in the villi being flattened.

Explain why some people with untreated coeliac disease will eventually develop osteoporosis.

**(4)**

OLevels.com SCHOOL

(c) Another condition of the alimentary canal is colon cancer. This is often treated by removal of the large intestine.

Explain why people who have this treatment often suffer from dehydration.

**(2)**

**(Total for Question 4 = 9 marks)**

## 6. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

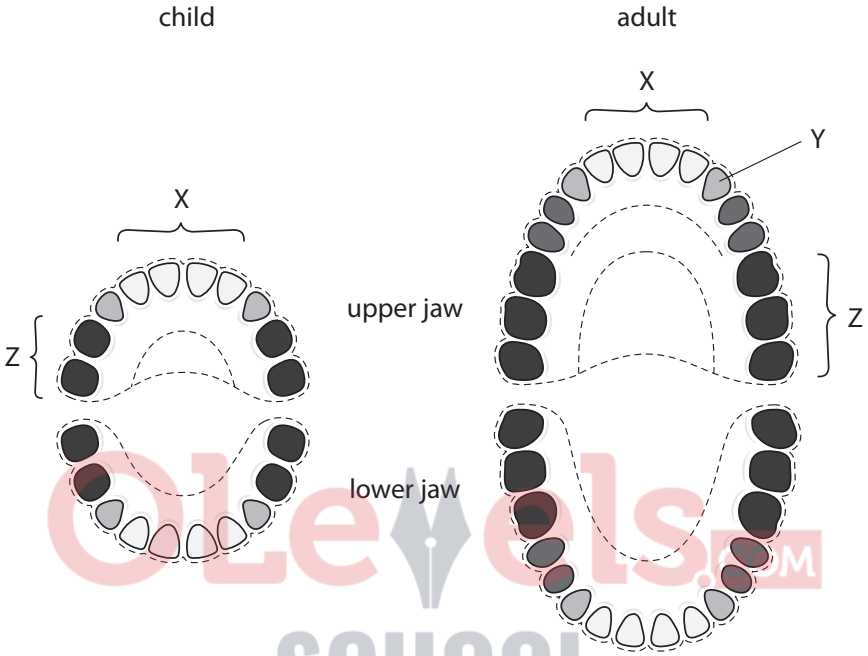
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# 7. Question 1 · part c.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c.i only.

**1** (a) The diagrams show the arrangement of teeth in a young child and the arrangement of teeth in an adult.




(Source: https://www.medicalnewstoday.com/articles/326754#number-of-teeth)

(i) Name the types of teeth shown by X and Z.

**(2)**

X .....

Z .....

(ii) State the function of the tooth labelled Y.

**(1)**

.....

.....

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DO NOT WRITE IN THIS AREA



P 7 4 4 9 5 A 0 2 2 8

# 7. Question 1 · part c.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c.i only.

DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA

(b) Digestion of food begins in the mouth.

Use the most appropriate words or numbers to complete the passage about digestion of food in the mouth.

**(4)**

Saliva in the mouth contains an enzyme called ..... .

This enzyme speeds up the digestion of starch to produce

..... .

The enzyme in saliva works best at pH ..... .

If the pH in the mouth was too low or too high, the digestion of carbohydrates

would ..... .

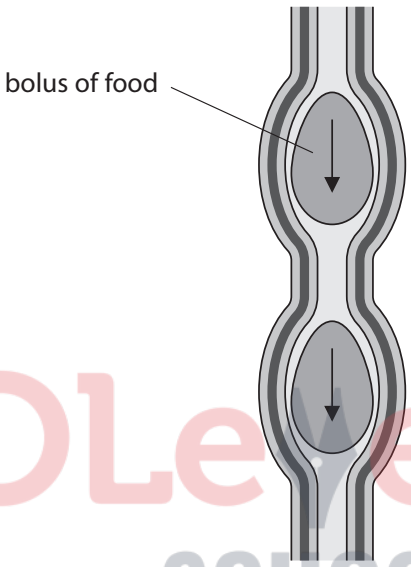
OLevels.com SCHOOL

# 7. Question 1 · part c.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c.i only.

(c) The diagram shows the part of the digestive system that transports food from the mouth to the stomach.




(Source: https://www.motifolio.com/1031265.html)

(i) Name the part of the digestive system that transports food from the mouth to the stomach.

**(1)**

.....

(ii) Describe how the action of this part of the digestive system transports food from the mouth to the stomach.

**(2)**

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.....

**(Total for Question 1 = 10 marks)**

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## 7. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

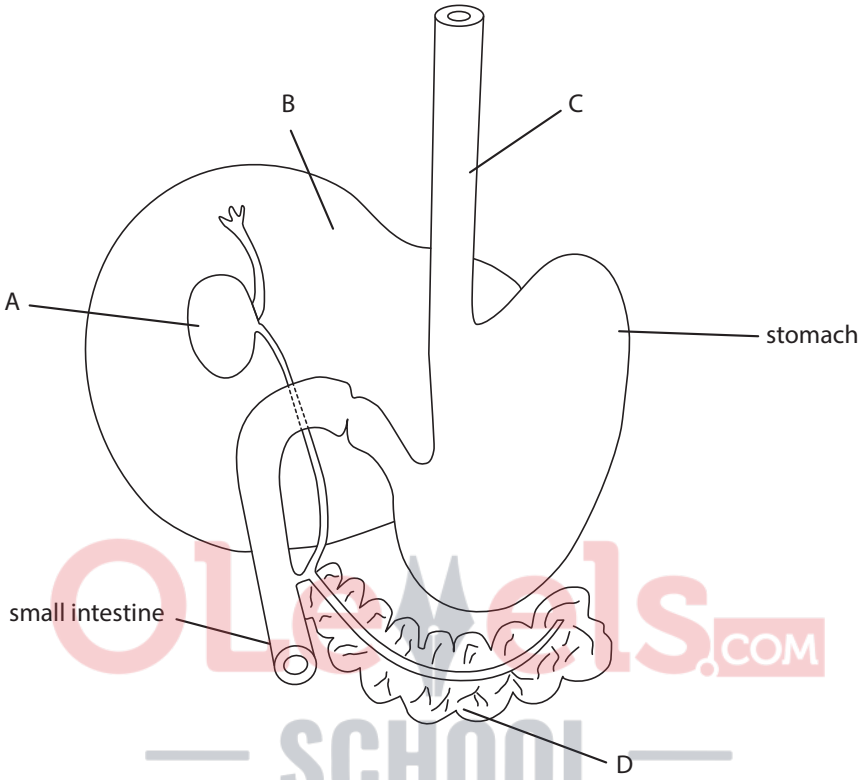
OLevels.com SCHOOL

# 8. Question 2 · part a.i · 4 marks

June 2025 Paper 2

Further practice · Attempt part a.i only.

**2** The diagram shows structures involved in digestion.




(a) (i) Identify the structures labelled A, B, C and D.

**(4)**

A .....

B .....

C .....

D .....

(ii) Name the fluid stored in structure A.

**(1)**

.....

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## 8. Question 2 · part a.i · 4 marks · Working and answer

Working and answer

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## 9. Question 4 · part b · 2 marks

January 2020 Paper 2

Further practice · Attempt part b only.

**4** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

**(1)**

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

**(2)**

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## 9. Question 4 · part b · 2 marks

January 2020 Paper 2

Further practice · Attempt part b only.

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(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

**(2)**

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

**(1)**

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

**(1)**

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

**(2)**

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

**(1)**

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## 9. Question 4 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 10. Question 1 · part b.ii · 4 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

**1** Food enters the alimentary canal through the mouth.

All undigested food leaves the body through the anus.

(a) Name three structures the food passes through between the mouth and the anus.

**(3)**

1 .....

2 .....

3 .....

(b) Some of the food is digested by enzymes as it passes through the alimentary canal.

(i) Name an enzyme that digests protein.

**(1)**

.....

(ii) Describe the processes that occur in the mouth to digest food.

**(4)**

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## 10. Question 1 · part b.ii · 4 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

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(c) Fibre in food cannot be digested.

Explain the importance of fibre in the diet.

**(3)**

**(Total for Question 1 = 11 marks)**

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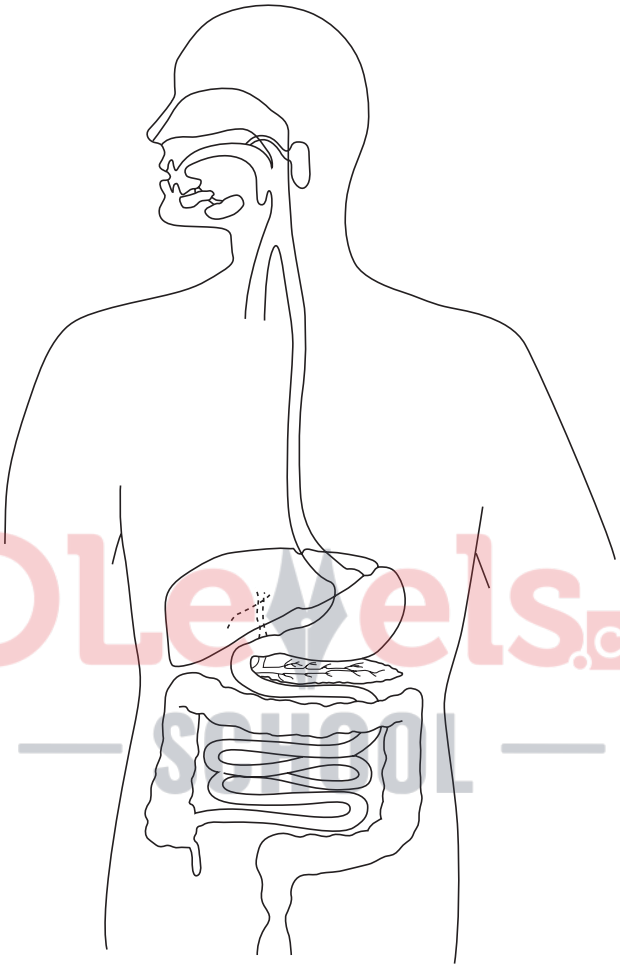
Working and answer

## 11. Question 4 · part c · 2 marks

November 2021 Paper 1

Further practice · Attempt part c only.

**4** The diagram shows the alimentary canal.



(a) Identify these areas of the alimentary canal using label lines and the correct letters.

(i) Label with an X the area where starch digestion begins.

**(1)**

(ii) Label with a Y the area that has a pH of 2

**(1)**

(iii) Label with a Z the area where both lipid and protein digesting enzymes are produced.

**(1)**

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P 6 7 0 6 0 R R A 0 8 2 0

## 11. Question 4 · part c · 2 marks

November 2021 Paper 1

Further practice · Attempt part c only.

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(b) Some people have the condition known as coeliac disease. This results in the villi being flattened.

Explain why some people with untreated coeliac disease will eventually develop osteoporosis.

**(4)**

OLevels.com SCHOOL

(c) Another condition of the alimentary canal is colon cancer. This is often treated by removal of the large intestine.

Explain why people who have this treatment often suffer from dehydration.

**(2)**

**(Total for Question 4 = 9 marks)**

## 11. Question 4 · part c · 2 marks · Working and answer

Working and answer

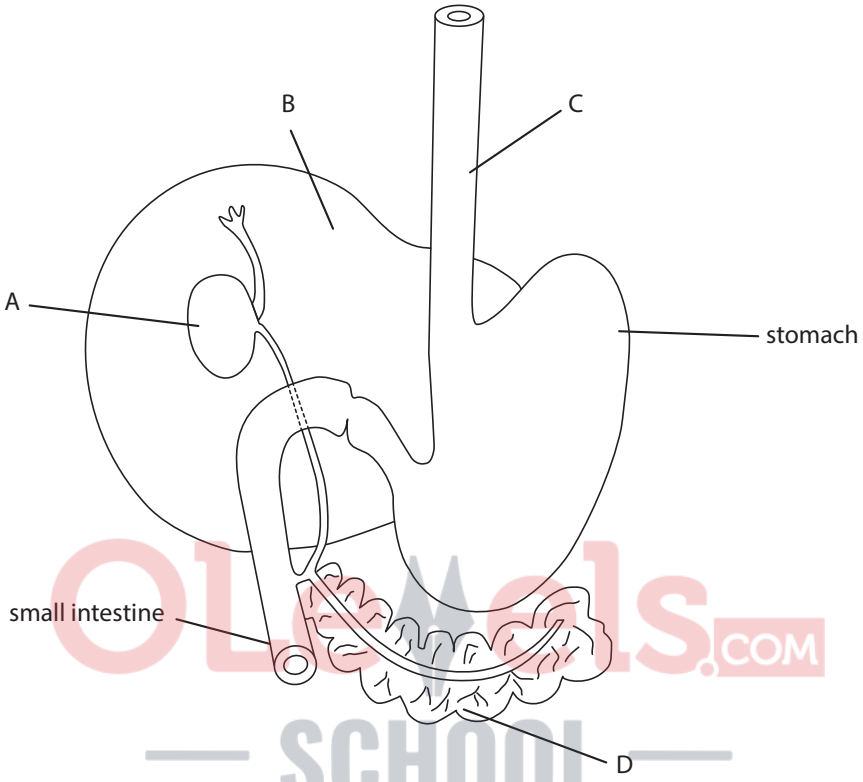
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# 12. Question 2 · part b.i · 4 marks

June 2025 Paper 2

Further practice · Attempt part b.i only.

**2** The diagram shows structures involved in digestion.




(a) (i) Identify the structures labelled A, B, C and D.

**(4)**

A .....

B .....

C .....

D .....

(ii) Name the fluid stored in structure A.

**(1)**

.....

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P 7 8 7 6 9 A 0 4 2 0

# 12. Question 2 · part b.i · 4 marks

June 2025 Paper 2

Further practice · Attempt part b.i only.

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(iii) Explain the function of the fluid stored in structure A in digestion.

**(3)**

(b) (i) Describe the role of structure D in digestion.

**(4)**

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(ii) Name two substances secreted by structure D that are not involved in digestion.

**(2)**

1 .....

2 .....

**(Total for Question 2 = 14 marks)**

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## 12. Question 2 · part b.i · 4 marks · Working and answer

Working and answer

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# Practice workbook · Link a nutrient to its function, source and deficiency disease

Functions and food sources of named nutrients and the deficiency diseases scurvy, anaemia, vitamin A blindness and kwashiorkor.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-NUTRIENT-FUNCTION-DEFICIENCY

# 1. Question 2 · part b · 2 marks

June 2019 Paper 1R

Worked-example retry · Attempt part b only.

**2** Humans need a balanced diet.

(a) Vitamins are needed in a balanced diet.

State four other components of a balanced diet.

**(2)**

1 .....

2 .....

3 .....

4 .....

(b) Explain why vitamin C is needed in a balanced diet.

**(2)**

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# 1. Question 2 · part b · 2 marks

June 2019 Paper 1R

Worked-example retry · Attempt part b only.

(c) The vitamin C content of fruit juices varies.

(i) Design an investigation to compare the vitamin C content of fresh fruit juice and long-life fruit juice.

**(5)**

(ii) State two variables that should be controlled in this investigation.

**(2)**

1 .....

2 .....

**(Total for Question 2 = 11 marks)**

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## 1. Question 2 · part b · 2 marks · Working and answer

Working and answer

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## 2. Question 4 · part c · 1 mark

June 2019 Paper 2

Further practice · Attempt part c only.

**4** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

In 1916, an American doctor, Joseph Goldberger, noticed symptoms of a mystery disease that were common among his patients. These symptoms included headaches, swollen tongues, skin rashes, upset stomachs and mental illness. In one American state, North Carolina, 40% of the 30 000 people living in this area died from this mystery disease.

Goldberger investigated whether the symptoms of the mystery disease were caused by pathogens. His investigation involved injecting himself with the blood from one patient, eating the skin rash of another and swallowing fluid taken from the intestines of a third. Goldberger did not get the disease, although he did suffer from nausea and diarrhoea.

Goldberger concluded that the symptoms shown by the patients were not caused by an infectious disease. He decided that they were more likely to be caused by the poor diet of the patients.

(a) Calculate the number of people in North Carolina who died from this mystery disease (lines 4 to 5).

**(2)**

number of deaths = .....

(b) State the meaning of the term pathogens (line 7).

**(1)**

.....

.....

(c) The people were suffering from a deficiency disease called pellagra, which is caused by a lack of vitamin B in the diet.

State why deficiency diseases such as pellagra are not infectious diseases.

**(1)**

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P 5 8 5 6 6 A 0 8 2 8

## 2. Question 4 · part c · 1 mark

June 2019 Paper 2

Further practice · Attempt part c only.

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(d) Describe a method that Goldberger could use to show that the symptoms of his patients were caused by a lack of vitamin B in their diet.

**(3)**

(e) Scurvy is another deficiency disease.

Scurvy affects the production of a protein called collagen.

Collagen is a component of many body tissues including cartilage.

Explain why a person with scurvy often experiences pain when walking.

**(2)**

**(Total for Question 4 = 9 marks)**

OLevels.COM SCHOOL

Working and answer

# 3. Question 1 · part b.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.i only.

**1** (a) Food packaging gives information about the food it contains.

The photograph shows some of the information given on a package containing yogurt.




(Source: © studiomode / Alamy Stock Photo)

The instructions state that the yogurt should be stored in a fridge at 4°C.

Why should this instruction be followed?

**(1)**

- ☒ **A** to make the yogurt taste better
- ☒ **B** to kill all bacteria in the yogurt
- ☒ **C** to slow down the growth of bacteria in the yogurt
- ☒ **D** to improve the appearance of the yogurt

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# 3. Question 1 · part b.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.i only.

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(b) The diagram shows the nutritional label on a package containing yogurt.

**Nutrition Facts**
Serving Size
1 container (150g)
Energy in kJ 480
Energy from Fat 0

| Amount/Serving | % Daily Value* | Amount/Serving | % Daily Value* |
|---|---|---|---|
| Total Fat | a% | Sodium | 2% |
| Saturated Fat | 0% | Total Carb. | 6% |
| Trans Fat | | Sugars | |
| Cholesterol | 1% | Protein | 22% |

Vitamin A 4% • Calcium 10% • Vitamin D 20%

Not a significant source of dietary fibre, vitamin C and iron.

(i) State which nutrient on the label is required for growth and repair of body tissue.

**(1)**

.....

(ii) Why is this yogurt suitable for a person who wants to reduce their body mass?

**(1)**

☒ **A** it contains vitamins C and D

☒ **B** it contains less than 5 mg of cholesterol

☒ **C** it does not contain energy from carbohydrates

☒ **D** it does not contain energy from fat

(c) Yogurt contains reducing sugars.

Describe a method to test for reducing sugars.

**(3)**

.....

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.....

**(Total for Question 1 = 6 marks)**

## 3. Question 1 · part b.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

# 4. Question 4 · part c.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part c.i only.

**4** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

**(1)**

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

**(2)**

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**8**

P 5 9 9 1 0 A 0 8 2 4

# 4. Question 4 · part c.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part c.i only.

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(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

**(2)**

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

**(1)**

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

**(1)**

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

**(2)**

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

**(1)**

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## 4. Question 4 · part c.i · 1 mark · Working and answer

Working and answer

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# 5. Question 4 · part c.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part c.ii only.

**4** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

**(1)**

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

**(2)**

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OLevels.com SCHOOL

**8**

P 5 9 9 1 0 A 0 8 2 4

## 5. Question 4 · part c.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

**(2)**

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

**(1)**

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

**(1)**

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

**(2)**

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

**(1)**

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## 5. Question 4 · part c.ii · 1 mark · Working and answer

Working and answer

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# 6. Question 4 · part a · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a only.

**4** (a) Kwashiorkor is a deficiency disease that results from the lack of a certain nutrient in the diet.

Use words from the box to complete the passage about kwashiorkor.

**(2)**

| fingers | carbohydrate | bread | ankles | protein | fat |
|---|---|---|---|---|---|

Kwashiorkor is a disease that is more common in children.

One symptom of kwashiorkor is swelling in the ..... .

The symptoms of kwashiorkor are corrected by adding more ..... to the diet.

(b) The table gives information about the number of cases of kwashiorkor recorded in a hospital over four years.

| Year | Number of cases of kwashiorkor |
|---|---|
| 2010 | 32 |
| 2011 | 21 |
| 2012 | 23 |
| 2013 | 19 |

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## 6. Question 4 · part a · 2 marks · Working and answer

Working and answer

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# 7. Question 4 · part b.ii · 1 mark

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

**4** (a) Kwashiorkor is a deficiency disease that results from the lack of a certain nutrient in the diet.

Use words from the box to complete the passage about kwashiorkor.

**(2)**

| fingers | carbohydrate | bread | ankles | protein | fat |
|---|---|---|---|---|---|

Kwashiorkor is a disease that is more common in children.

One symptom of kwashiorkor is swelling in the ..... .

The symptoms of kwashiorkor are corrected by adding more ..... to the diet.

(b) The table gives information about the number of cases of kwashiorkor recorded in a hospital over four years.

| Year | Number of cases of kwashiorkor |
|---|---|
| 2010 | 32 |
| 2011 | 21 |
| 2012 | 23 |
| 2013 | 19 |

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## 7. Question 4 · part b.ii · 1 mark

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

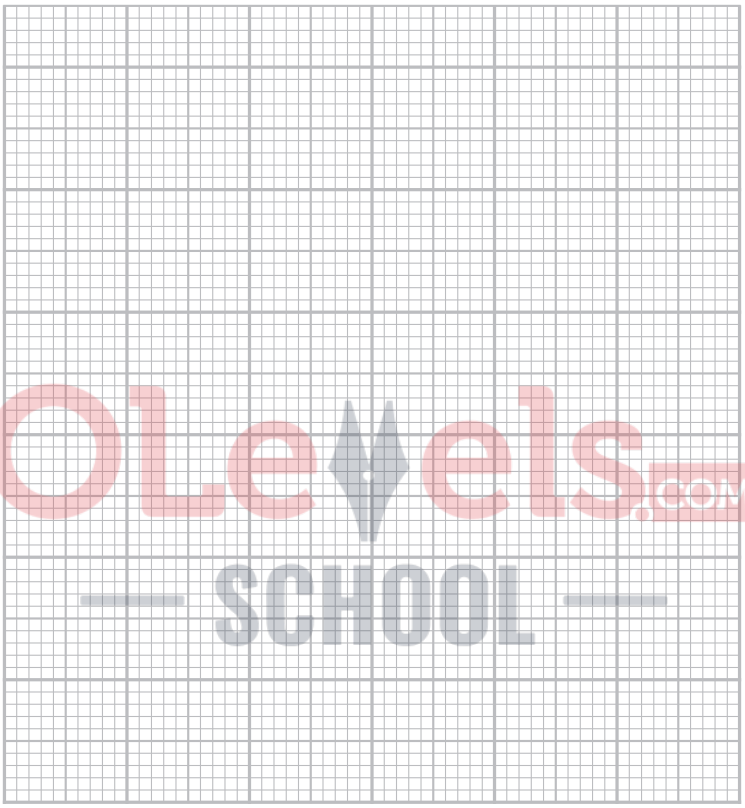
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(i) Plot a bar chart of the data on the grid.

**(3)**



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(ii) Suggest a reason for the difference in the number of recorded cases of kwashiorkor in 2010 compared with 2013.

**(1)**

.....

.....

(iii) In a hospital, 95 out of a total of 1978 children with malnutrition were found to have kwashiorkor.

Calculate the percentage of children with kwashiorkor.

**(2)**

percentage of children = ..... %

**(Total for Question 4 = 8 marks)**

## 7. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

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## 8. Question 5 · part a.i · 1 mark

November 2020 Paper 1R

Further practice · Attempt part a.i only.

**5** (a) The type and amount of different nutrients needed for a balanced diet varies depending on several factors.

(i) Pregnant women are often advised by their doctor to take extra iron in their diet.

Which two foods are the best sources of iron?

**(1)**

- ☒ **A** fruit and green vegetables
- ☒ **B** dairy products and red meat
- ☒ **C** green vegetables and red meat
- ☒ **D** fruit and dairy products

(ii) Explain why a woman should have more iron in her diet when she becomes pregnant.

**(4)**

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## 8. Question 5 · part a.i · 1 mark · Working and answer

Working and answer

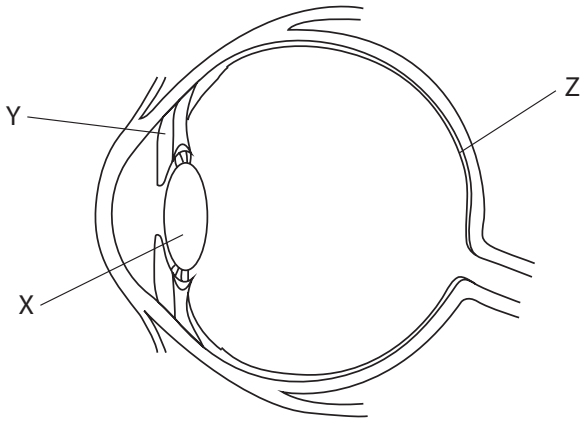
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# 9. Question 1 · part b.iv · 1 mark

January 2023 Paper 1

Further practice · Attempt part b.iv only.

**1** (a) The diagram shows a human eye.




(i) Draw one line from each structure to its correct function.

**(3)**

| Structure | Function |
|---|---|
| structure X | refracts light rays |
| structure Y | protects the eye from pathogens |
| structure Z | detects light rays |
| | controls the shape of the lens |
| | transmits electrical impulses to the brain |
| | controls the size of the pupil |

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# 9. Question 1 · part b.iv · 1 mark

January 2023 Paper 1

Further practice · Attempt part b.iv only.

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(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

**(1)**

- ☒ **A** being able to judge depth
- ☒ **B** being able to see in colour
- ☒ **C** having a wide field of view
- ☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.



Healthy eye Eye with cataract

(i) Describe the cause of cataracts.

**(2)**

.....

.....

.....

(ii) State one treatment for cataracts.

**(1)**

.....

(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

**(2)**

number of people = .....

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

**(1)**

.....

**(Total for Question 1 = 10 marks)**

## 9. Question 1 · part b.iv · 1 mark · Working and answer

Working and answer

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## 10. Question 4 · part d · 5 marks

June 2023 Paper 1R

Further practice · Attempt part d only.

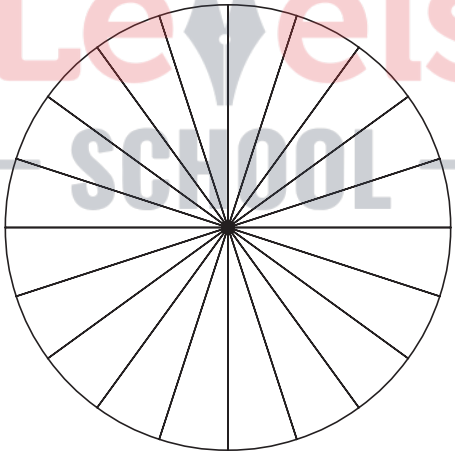
**4** The table shows the percentage composition of the diets of two 10 year-old girls.

One girl lives in Area A and the other girl lives in Area B.

| Component | Percentage (%) composition of diet | |
|---|---|---|
| | Area A | Area B |
| cereals | 75.0 | 15.0 |
| fruit and vegetables | 15.0 | 37.5 |
| eggs, fish and meat | 2.5 | 27.5 |
| milk, cheese and butter | 7.5 | 20.0 |

(a) Complete the pie chart to show the diet of the girl in Area B.

**(4)**



(b) Explain what is meant by a balanced diet.

**(2)**

.....

.....

.....

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## 10. Question 4 · part d · 5 marks

June 2023 Paper 1R

Further practice · Attempt part d only.

(c) Calculate the ratio of the percentage composition of cereals in the diet of the girl in Area A compared with the diet of the girl in Area B.

Give your answer in the form n : 1

**(2)**

ratio = .....

(d) Discuss the effects that the percentage composition given for eggs, fish and meat in the diet may have on the growth of the two girls.

**(5)**

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**(Total for Question 4 = 13 marks)**

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## 10. Question 4 · part d · 5 marks · Working and answer

Working and answer

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# 11. Question 3 · part a.iv · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.iv only.

**3** (a) The table gives the recommended daily dietary intake for different groups of males and females.

| Group | Energy in kJ | Protein in g | Vitamin A in mg | Vitamin C in mg | Iron in mg |
|---|---|---|---|---|---|
| MALES | | | | | |
| 15 to 18 years old | 12 600 | 59 | 0.9 | 50 | 12 |
| 25 to 50 years old | 12 200 | 63 | 0.9 | 60 | 10 |
| FEMALES | | | | | |
| 15 to 18 years old | 9200 | 46 | 0.7 | 50 | 15 |
| 25 to 50 years old | 9200 | 50 | 0.7 | 60 | 15 |

(i) Describe what is meant by a balanced diet.

**(2)**

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

**(3)**

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# 11. Question 3 · part a.iv · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.iv only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

**(2)**

.....

.....

.....

.....

(iv) State a nutrient, shown in the table, that can help to reduce night blindness.

**(1)**

.....

(v) Red meat and some green vegetables are good sources of iron.

100 g of red meat contains 2.5 mg of iron.

100 g of broccoli contains 0.70 mg of iron.

A male in the age range 15 to 18 years eats 250 g of red meat and 200 g of broccoli in one meal.

Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

**(4)**

extra mass = ..... mg

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## 11. Question 3 · part a.iv · 1 mark · Working and answer

Working and answer

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## 12. Question 3 · part b.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.i only.

**3** (a) The table gives the recommended daily dietary intake for different groups of males and females.

| Group | Energy in kJ | Protein in g | Vitamin A in mg | Vitamin C in mg | Iron in mg |
|---|---|---|---|---|---|
| MALES | | | | | |
| 15 to 18 years old | 12 600 | 59 | 0.9 | 50 | 12 |
| 25 to 50 years old | 12 200 | 63 | 0.9 | 60 | 10 |
| FEMALES | | | | | |
| 15 to 18 years old | 9200 | 46 | 0.7 | 50 | 15 |
| 25 to 50 years old | 9200 | 50 | 0.7 | 60 | 15 |

(i) Describe what is meant by a balanced diet.

**(2)**

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

**(3)**

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## 12. Question 3 · part b.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.i only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

**(2)**

.....

.....

.....

.....

(iv) State a nutrient, shown in the table, that can help to reduce night blindness.

**(1)**

.....

(v) Red meat and some green vegetables are good sources of iron.

100 g of red meat contains 2.5 mg of iron.

100 g of broccoli contains 0.70 mg of iron.

A male in the age range 15 to 18 years eats 250 g of red meat and 200 g of broccoli in one meal.

Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

**(4)**

extra mass = ..... mg

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## 12. Question 3 · part b.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.i only.

(b) In the 19th century more than two million sailors on long ocean voyages died from a nutritional disease called scurvy.

(i) Give a symptom of scurvy.

**(1)**

.....

.....

(ii) Suggest why scurvy is an uncommon nutritional disease in the modern world.

**(1)**

.....

.....

**(Total for Question 3 = 14 marks)**

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Working and answer

## 13. Question 3 · part b.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

**3** (a) The table gives the recommended daily dietary intake for different groups of males and females.

| Group | Energy in kJ | Protein in g | Vitamin A in mg | Vitamin C in mg | Iron in mg |
|---|---|---|---|---|---|
| MALES | | | | | |
| 15 to 18 years old | 12 600 | 59 | 0.9 | 50 | 12 |
| 25 to 50 years old | 12 200 | 63 | 0.9 | 60 | 10 |
| FEMALES | | | | | |
| 15 to 18 years old | 9200 | 46 | 0.7 | 50 | 15 |
| 25 to 50 years old | 9200 | 50 | 0.7 | 60 | 15 |

(i) Describe what is meant by a balanced diet.

**(2)**

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

**(3)**

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# 13. Question 3 · part b.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

**(2)**

(iv) State a nutrient, shown in the table, that can help to reduce night blindness.

**(1)**

(v) Red meat and some green vegetables are good sources of iron.

100 g of red meat contains 2.5 mg of iron.

100 g of broccoli contains 0.70 mg of iron.

A male in the age range 15 to 18 years eats 250 g of red meat and 200 g of broccoli in one meal.

Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

**(4)**

extra mass = ..... mg

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## 13. Question 3 · part b.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

(b) In the 19th century more than two million sailors on long ocean voyages died from a nutritional disease called scurvy.

(i) Give a symptom of scurvy.

**(1)**

.....

.....

(ii) Suggest why scurvy is an uncommon nutritional disease in the modern world.

**(1)**

.....

.....

**(Total for Question 3 = 14 marks)**

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Working and answer

Practice workbook · Name tooth types and tissues, state their jobs and compare tooth sets

Incisor, canine, premolar and molar, enamel, dentine, pulp and nerve, the job of each tooth type, and differences between milk and adult sets.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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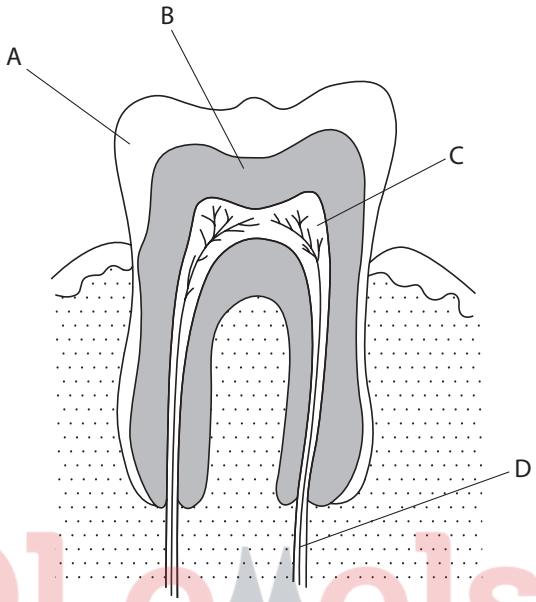
Type ID: HBI-TOOTH-TYPES-PARTS-AND-FUNCTIONS

# 1. Question 3 · part a.ii · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part a.ii only.

**3** (a) The diagram shows a tooth.




(i) Name the type of tooth.

**(1)**

.....

(ii) Identify the parts of the tooth labelled A, B, C and D.

**(4)**

A .....

B .....

C .....

D .....

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# 1. Question 3 · part a.ii · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part a.ii only.

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(iii) Explain why this type of tooth is more likely to have tooth decay than other types of teeth.

**(3)**

(b) Explain why the minerals obtained from a balanced diet are important in developing and maintaining healthy teeth.

**(4)**

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(c) State three other ways to keep teeth healthy.

**(3)**

1 .....

2 .....

3 .....

**(Total for Question 3 = 15 marks)**

## 1. Question 3 · part a.ii · 4 marks · Working and answer

Working and answer

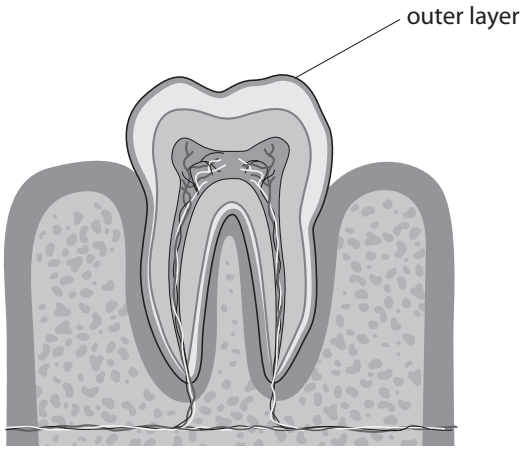
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## 2. Question 1 · part a.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.i only.

**1** (a) The diagram shows a section through one type of tooth.




(i) State the name of this type of tooth.

**(1)**

.....

(ii) Which substance forms the outer layer of the tooth?

**(1)**

- ☒ **A** bone
- ☒ **B** dentine
- ☒ **C** enamel
- ☒ **D** pulp

(iii) Explain why this type of tooth is more likely to decay than other types of teeth.

**(3)**

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## 2. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

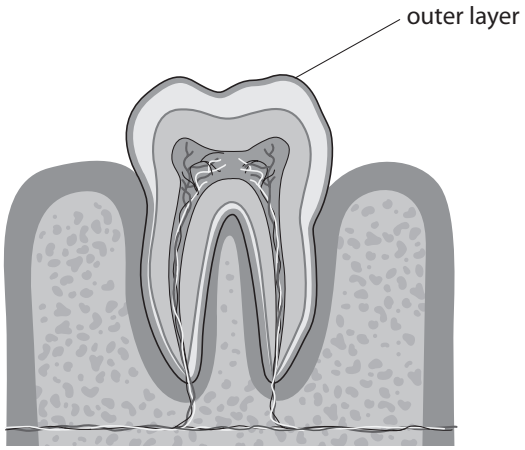
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# 3. Question 1 · part a.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.ii only.

**1** (a) The diagram shows a section through one type of tooth.




(i) State the name of this type of tooth.

**(1)**

.....

(ii) Which substance forms the outer layer of the tooth?

**(1)**

- ☒ **A** bone
- ☒ **B** dentine
- ☒ **C** enamel
- ☒ **D** pulp

(iii) Explain why this type of tooth is more likely to decay than other types of teeth.

**(3)**

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## 3. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

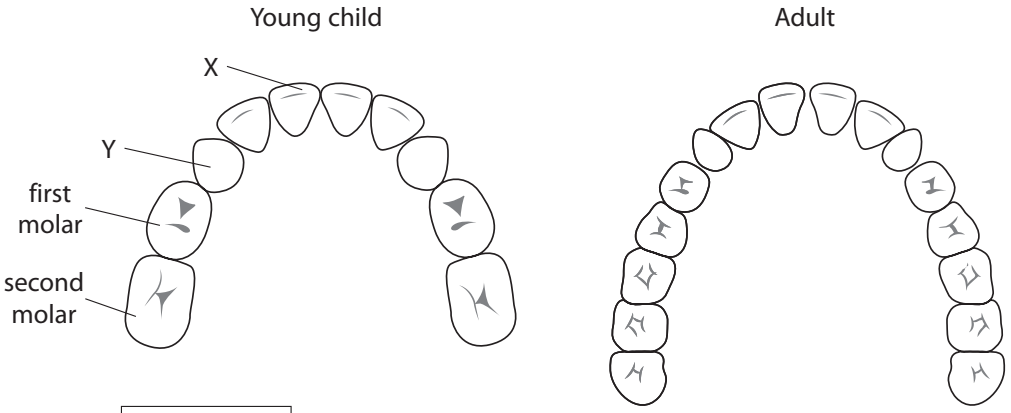
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# 4. Question 1 · part a.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

**1** The diagram shows the upper teeth of a young child and the upper teeth of an adult.




not to scale

(a) (i) Name tooth X and tooth Y. **(2)**

tooth X .....

tooth Y .....

(ii) State the function of the molar teeth. **(1)**

.....

.....

(iii) State two differences between the teeth of a young child and the teeth of an adult. **(2)**

1 .....

.....

2 .....

.....

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## 4. Question 1 · part a.i · 2 marks · Working and answer

Working and answer

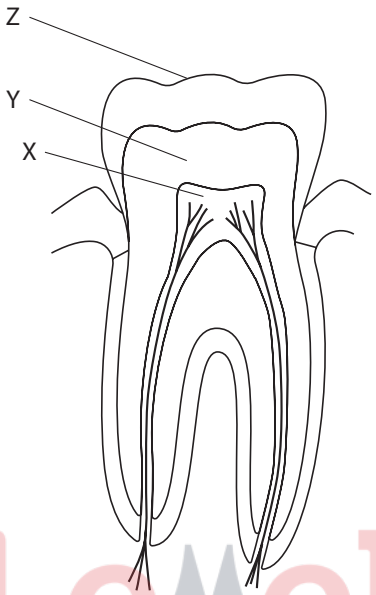
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# 5. Question 7 · part a.i · 3 marks

June 2023 Paper 2

Further practice · Attempt part a.i only.

**7** The diagram shows a section through a tooth.




(a) (i) Name the parts of the tooth labelled X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Explain where decay in the tooth is most likely to occur.

**(3)**

.....

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**20**

## 5. Question 7 · part a.i · 3 marks · Working and answer

Working and answer

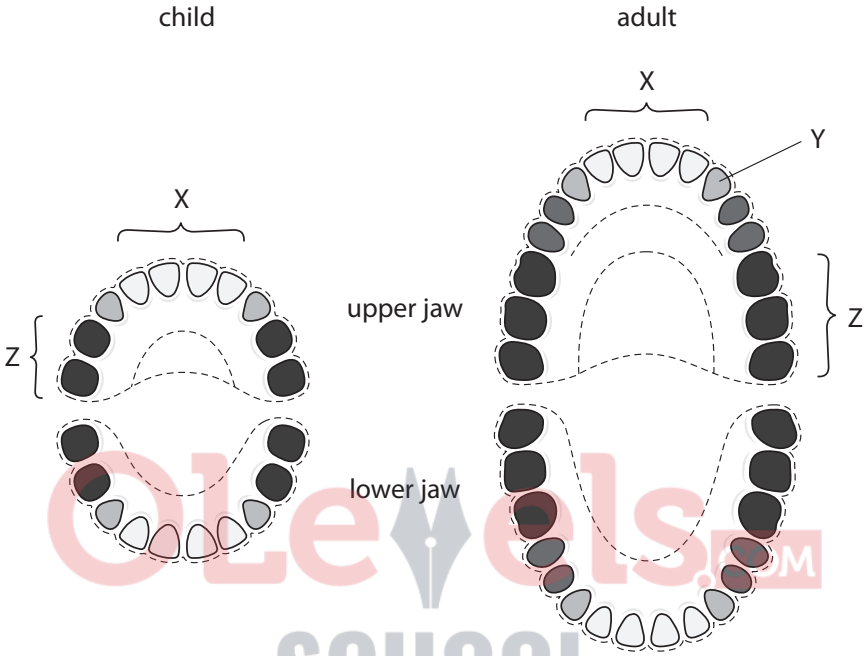
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# 6. Question 1 · part a.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part a.i only.

**1** (a) The diagrams show the arrangement of teeth in a young child and the arrangement of teeth in an adult.




(Source: https://www.medicalnewstoday.com/articles/326754#number-of-teeth)

(i) Name the types of teeth shown by X and Z.

**(2)**

X .....

Z .....

(ii) State the function of the tooth labelled Y.

**(1)**

.....

.....

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## 6. Question 1 · part a.i · 2 marks · Working and answer

Working and answer

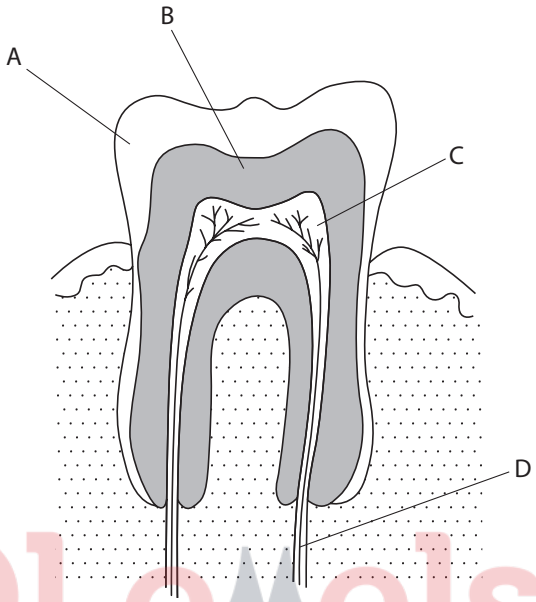
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# 7. Question 3 · part a.i · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.i only.

**3** (a) The diagram shows a tooth.




(i) Name the type of tooth.

**(1)**

.....

(ii) Identify the parts of the tooth labelled A, B, C and D.

**(4)**

A .....

B .....

C .....

D .....

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## 7. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

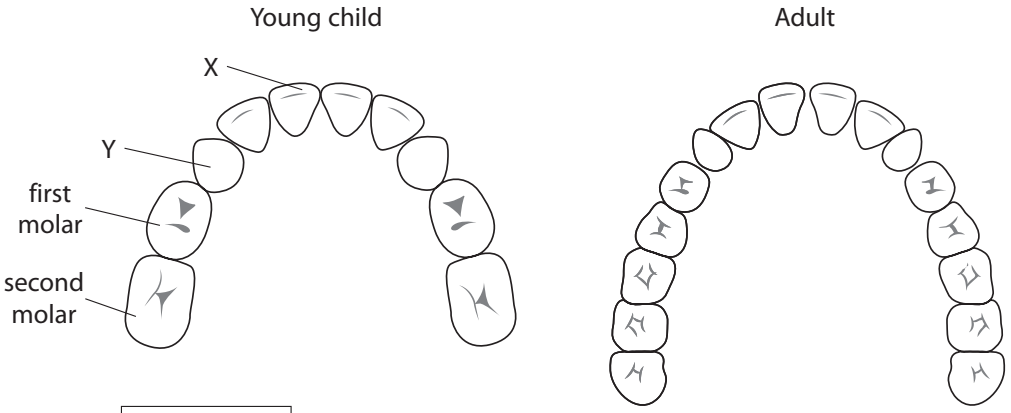
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## 8. Question 1 · part a.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part a.ii only.

**1** The diagram shows the upper teeth of a young child and the upper teeth of an adult.




not to scale

(a) (i) Name tooth X and tooth Y. **(2)**

tooth X .....

tooth Y .....

(ii) State the function of the molar teeth. **(1)**

.....

.....

(iii) State two differences between the teeth of a young child and the teeth of an adult. **(2)**

1 .....

.....

2 .....

.....

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## 8. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

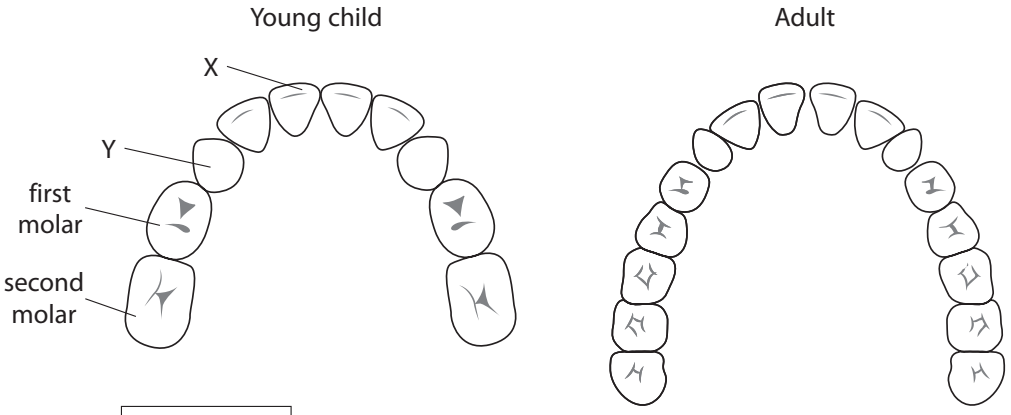
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## 9. Question 1 · part a.iii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iii only.

**1** The diagram shows the upper teeth of a young child and the upper teeth of an adult.




not to scale

(a) (i) Name tooth X and tooth Y. **(2)**

tooth X .....

tooth Y .....

(ii) State the function of the molar teeth. **(1)**

.....

.....

(iii) State two differences between the teeth of a young child and the teeth of an adult. **(2)**

1 .....

.....

2 .....

.....

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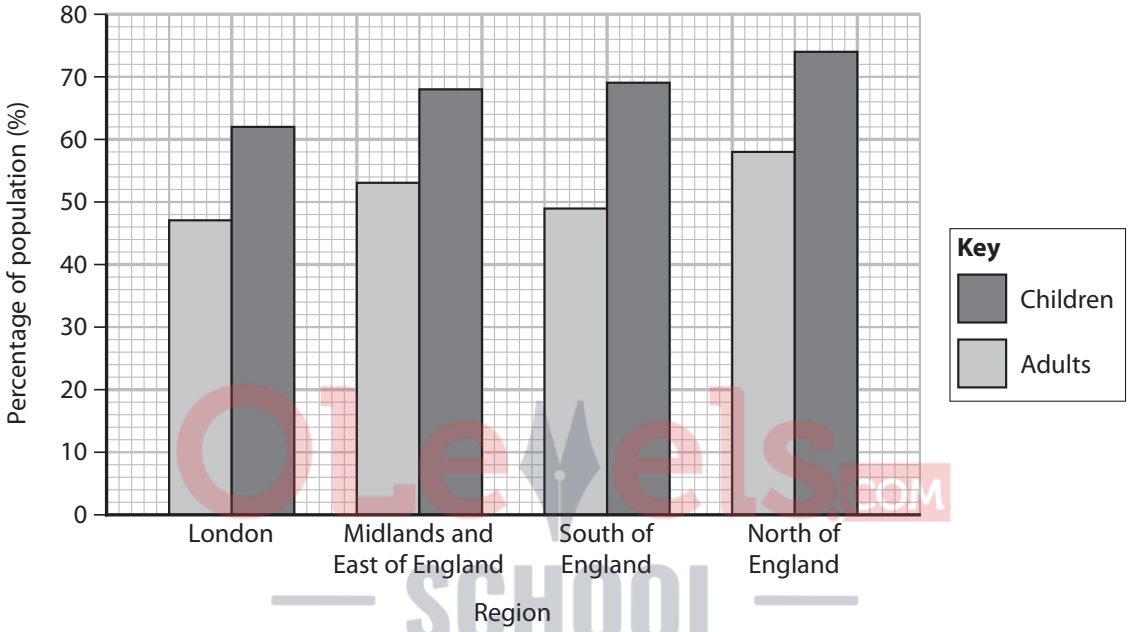
P 5 9 9 1 0 A 0 2 2 4

## 9. Question 1 · part a.iii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iii only.

(b) The graph shows the percentage of adults and children in different regions of England who visited a dentist in a two-year period.




Data source/s: Health and Social Care Information Centre NHS Dental Activity Statistics: December 2013 and GP Patient Survey Dental Results 2013/14

(i) Explain which region is likely to have more adults and children with tooth decay.

**(2)**

.....

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(ii) Give two reasons why molars are more likely to be damaged by tooth decay than other teeth.

**(2)**

1 .....

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2 .....

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**(Total for Question 1 = 9 marks)**

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## 9. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

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## 10. Question 4 · part b · 2 marks

November 2021 Paper 2

Further practice · Attempt part b only.

**4** The table shows the results of a survey about the number of teeth per student in a class of young students.

| Number of teeth | Number of students |
|---|---|
| 23 | 1 |
| 24 | 2 |
| 25 | 3 |
| 26 | 4 |
| 27 | 7 |
| 28 | 13 |

(a) (i) Draw a bar chart to show the results of the survey.

**(4)**



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## 10. Question 4 · part b · 2 marks

November 2021 Paper 2

Further practice · Attempt part b only.

(ii) A full set of teeth for the students is 28

Calculate the percentage of the students in the class who have a full set of teeth.

**(3)**

percentage = ..... %

(b) A full set of teeth for an adult is 32

Suggest why the full set of teeth for these students is only 28

**(2)**

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## 10. Question 4 · part b · 2 marks

November 2021 Paper 2

Further practice · Attempt part b only.

(c) (i) Explain how reducing the amount of sugar in foods helps prevent tooth decay.

**(4)**

(ii) State two other methods that could reduce tooth decay.

**(2)**

1 .....

2 .....

**(Total for Question 4 = 15 marks)**

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## 10. Question 4 · part b · 2 marks · Working and answer

Working and answer

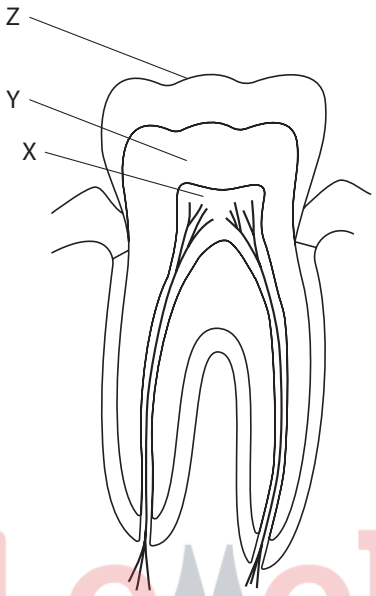
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# 11. Question 7 · part c · 3 marks

June 2023 Paper 2

Further practice · Attempt part c only.

**7** The diagram shows a section through a tooth.




(a) (i) Name the parts of the tooth labelled X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Explain where decay in the tooth is most likely to occur.

**(3)**

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**20**

P 7 2 5 8 7 A 0 2 0 2 4

## 11. Question 7 · part c · 3 marks

June 2023 Paper 2

Further practice · Attempt part c only.

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(b) Explain the role of bacteria in the process of tooth decay.

**(4)**

DO NOT WRITE IN THIS AREA

(c) Babies are usually born without teeth. The teeth develop many months after birth.

Explain why the teeth do not develop until many months after birth.

**(3)**

DO NOT WRITE IN THIS AREA

(d) Dentists recommend using a mouthwash morning and night to reduce dental decay.

One type of mouthwash contains alcohol and sodium fluoride.

Explain how this mouthwash could help to reduce dental decay.

**(2)**

**(Total for Question 7 = 15 marks)**

## 11. Question 7 · part c · 3 marks · Working and answer

Working and answer

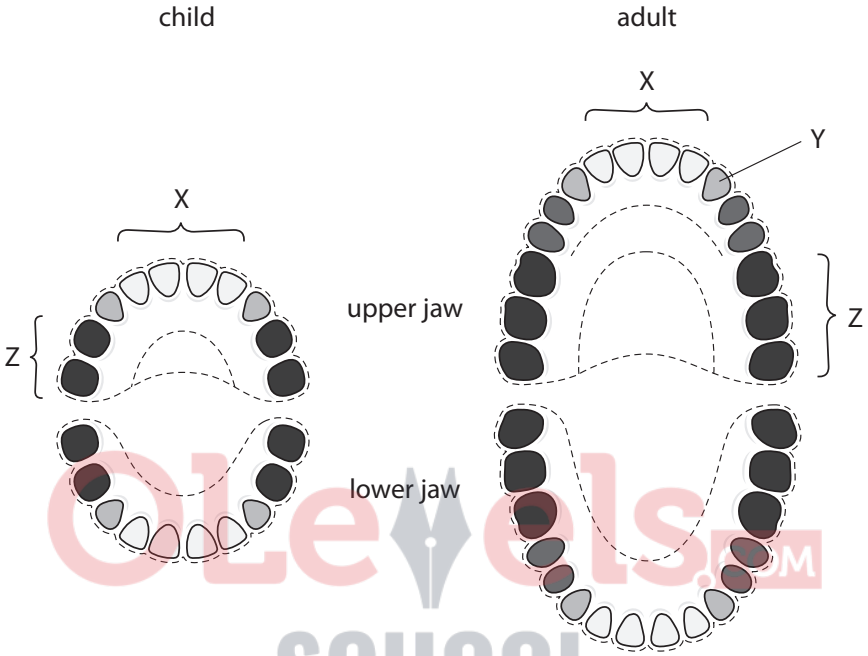
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# 12. Question 1 · part a.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.ii only.

**1** (a) The diagrams show the arrangement of teeth in a young child and the arrangement of teeth in an adult.




(Source: https://www.medicalnewstoday.com/articles/326754#number-of-teeth)

(i) Name the types of teeth shown by X and Z.

**(2)**

X .....

Z .....

(ii) State the function of the tooth labelled Y.

**(1)**

.....

.....

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## 12. Question 1 · part a.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.ii only.

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(b) Digestion of food begins in the mouth.

Use the most appropriate words or numbers to complete the passage about digestion of food in the mouth.

**(4)**

Saliva in the mouth contains an enzyme called ..... .

This enzyme speeds up the digestion of starch to produce

..... .

The enzyme in saliva works best at pH ..... .

If the pH in the mouth was too low or too high, the digestion of carbohydrates

would ..... .

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## 12. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Define a balanced diet and explain why needs differ between people

Meaning of a balanced diet and its components, and differences in energy or nutrients with sex, age, pregnancy, menstruation or infancy.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-BALANCED-DIET-AND-DIFFERING-NEEDS

# 1. Question 2 · part a · 2 marks

November 2024 Paper 2

Worked-example retry · Attempt part a only.

**2** (a) Describe what is meant by a balanced diet.

**(2)**

(b) A label on a pack of four slices of roast beef gives information about how much of each dietary component is present in the meat.

It also gives the percentage of the recommended daily amount for each dietary component.

Table 1 and table 2 give these values.

| Energy in kJ | Fat in g | Sugars in g | Salt in g |
|---|---|---|---|
| 470 | 2.4 | less than 0.4 | 0.6 |

**Table 1**

| Percentage of recommended daily amount (%) | | | |
|---|---|---|---|
| **energy** | **fat** | **sugars** | **salt** |
| 4.0 | 4.0 | less than 1.0 | 12.0 |

**Table 2**

(i) Name another dietary component that could be listed on the pack of meat.

**(1)**

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## 1. Question 2 · part a · 2 marks · Working and answer

Working and answer

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# 2. Question 2 · part a · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

**2** Humans need a balanced diet.

(a) Vitamins are needed in a balanced diet.

State four other components of a balanced diet.

**(2)**

1 .....

2 .....

3 .....

4 .....

(b) Explain why vitamin C is needed in a balanced diet.

**(2)**

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## 2. Question 2 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 3. Question 4 · part b · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b only.

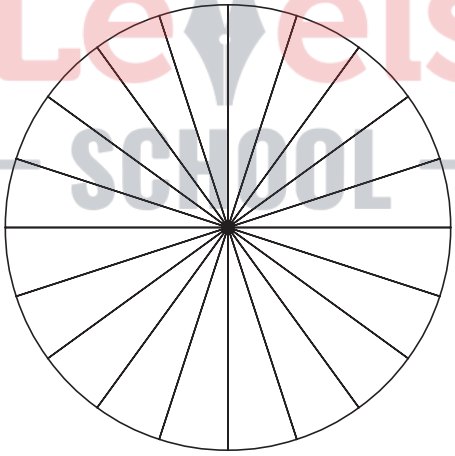
**4** The table shows the percentage composition of the diets of two 10 year-old girls.

One girl lives in Area A and the other girl lives in Area B.

| Component | Percentage (%) composition of diet | |
|---|---|---|
| | Area A | Area B |
| cereals | 75.0 | 15.0 |
| fruit and vegetables | 15.0 | 37.5 |
| eggs, fish and meat | 2.5 | 27.5 |
| milk, cheese and butter | 7.5 | 20.0 |

(a) Complete the pie chart to show the diet of the girl in Area B.

**(4)**



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(b) Explain what is meant by a balanced diet.

**(2)**

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## 3. Question 4 · part b · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b only.

(c) Calculate the ratio of the percentage composition of cereals in the diet of the girl in Area A compared with the diet of the girl in Area B.

Give your answer in the form n : 1

**(2)**

ratio = .....

(d) Discuss the effects that the percentage composition given for eggs, fish and meat in the diet may have on the growth of the two girls.

**(5)**

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**(Total for Question 4 = 13 marks)**

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## 3. Question 4 · part b · 2 marks · Working and answer

Working and answer

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# 4. Question 3 · part a.i · 2 marks

June 2024 Paper 1R

Further practice · Attempt part a.i only.

**3** (a) The table gives the recommended daily dietary intake for different groups of males and females.

| Group | Energy in kJ | Protein in g | Vitamin A in mg | Vitamin C in mg | Iron in mg |
|---|---|---|---|---|---|
| MALES | | | | | |
| 15 to 18 years old | 12 600 | 59 | 0.9 | 50 | 12 |
| 25 to 50 years old | 12 200 | 63 | 0.9 | 60 | 10 |
| FEMALES | | | | | |
| 15 to 18 years old | 9200 | 46 | 0.7 | 50 | 15 |
| 25 to 50 years old | 9200 | 50 | 0.7 | 60 | 15 |

(i) Describe what is meant by a balanced diet.

**(2)**

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

**(3)**

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P 7 4 4 9 2 A 0 8 2 4

## 4. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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# 5. Question 2 · part b.i · 1 mark

November 2024 Paper 2

Further practice · Attempt part b.i only.

**2** (a) Describe what is meant by a balanced diet.

**(2)**

(b) A label on a pack of four slices of roast beef gives information about how much of each dietary component is present in the meat.

It also gives the percentage of the recommended daily amount for each dietary component.

Table 1 and table 2 give these values.

| Energy in kJ | Fat in g | Sugars in g | Salt in g |
|---|---|---|---|
| 470 | 2.4 | less than 0.4 | 0.6 |

**Table 1**

| Percentage of recommended daily amount (%) | | | |
|---|---|---|---|
| **energy** | **fat** | **sugars** | **salt** |
| 4.0 | 4.0 | less than 1.0 | 12.0 |

**Table 2**

(i) Name another dietary component that could be listed on the pack of meat.

**(1)**

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P 7 5 9 5 9 A 0 4 2 4

## 5. Question 2 · part b.i · 1 mark

November 2024 Paper 2

Further practice · Attempt part b.i only.

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(ii) Using data from the tables, calculate the mass of fat, in g, that is needed to achieve the recommended daily amount.

**(3)**

mass = ..... g

(iii) Calculate the number of slices of meat a person would need to eat to supply a recommended daily energy intake of 11 800 kJ.

**(2)**

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number of slices = .....



P 7 5 9 5 9 A 0 5 2 4

*Turn over* ▶

## 5. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

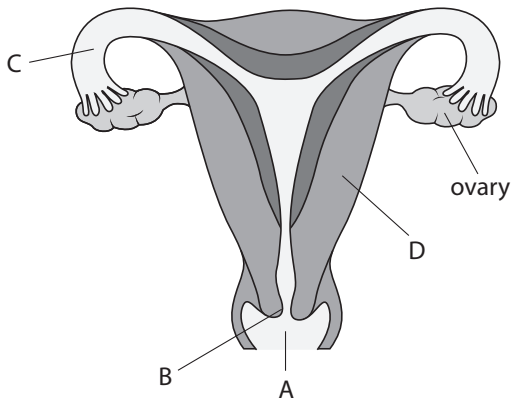
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# 6. Question 2 · part b · 1 mark

January 2020 Paper 1

Further practice · Attempt part b only.

**2** (a) The diagram shows some structures of the female reproductive system.




(i) Where are sperm deposited during sexual intercourse?

**(1)**

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(ii) Which part of the female reproductive system contracts during birth to push the fetus downwards?

**(1)**

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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## 6. Question 2 · part b · 1 mark

January 2020 Paper 1

Further practice · Attempt part b only.

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(iii) The ovaries produce female sex hormones.

The box lists words associated with the female reproductive system.

| | | | | |
|---|---|---|---|---|
| FSH | wall | fertilisation | oestrogen | ovulation |
| lining | ADH | follicles | menstruation | |

Use words from the box to complete the passage about the ovaries.

**(3)**

The ovaries produce a hormone called ..... .

This hormone repairs the uterus ..... following menstruation. Once a month, one egg is released from the ovaries. This is known as ..... .

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(b) State why a pregnant female should have more calcium in her diet than a female who is not pregnant.

**(1)**

.....

.....

(c) Some women breastfeed their babies.

Give one advantage of breastfeeding babies rather than feeding them bottled milk.

**(1)**

.....

.....

## 6. Question 2 · part b · 1 mark · Working and answer

Working and answer

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## 7. Question 5 · part a.ii · 4 marks

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

**5** (a) The type and amount of different nutrients needed for a balanced diet varies depending on several factors.

(i) Pregnant women are often advised by their doctor to take extra iron in their diet.

Which two foods are the best sources of iron?

**(1)**

- ☒ **A** fruit and green vegetables
- ☒ **B** dairy products and red meat
- ☒ **C** green vegetables and red meat
- ☒ **D** fruit and dairy products

(ii) Explain why a woman should have more iron in her diet when she becomes pregnant.

**(4)**

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# 7. Question 5 · part a.ii · 4 marks

November 2020 Paper 1R

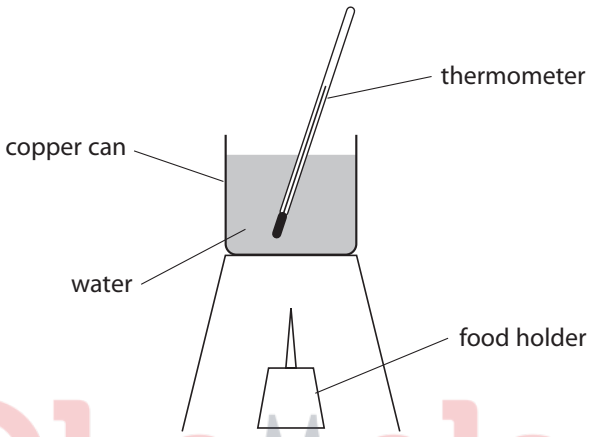
Further practice · Attempt part a.ii only.

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(b) The diagram shows some apparatus that can be used to measure the amount of energy contained in different foods.




(i) Explain how the apparatus can be used to obtain data about the energy content of different foods.

**(4)**

(ii) Explain how this apparatus can be changed to improve the accuracy of the data collected for each food.

**(3)**

**(Total for Question 5 = 12 marks)**

## 7. Question 5 · part a.ii · 4 marks · Working and answer

Working and answer

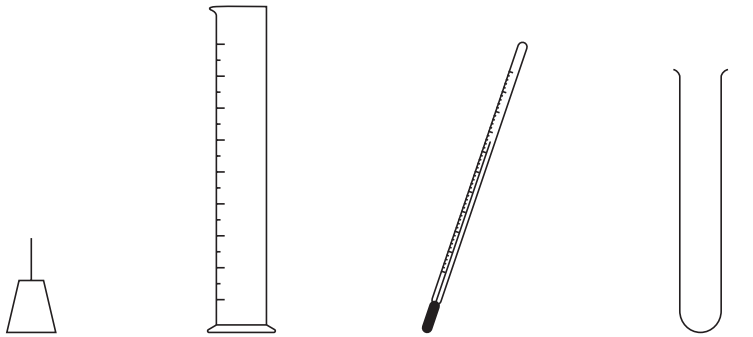
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# 8. Question 3 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

**3** (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

**(5)**

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# 8. Question 3 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Name two variables that need to be controlled in the investigation.

**(2)**

1 .....

.....

2 .....

.....

(iii) Give two safety precautions needed for this investigation.

**(2)**

1 .....

.....

2 .....

.....

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# 8. Question 3 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

(b) Different people have different energy requirements.

The table shows the daily energy requirements in kilocalories (kcal) of males and females of different ages.

**Daily Energy Requirement in kcal**

| Age of adults in years | Males | Females |
|---|---|---|
| 19–24 | 2772 | 2175 |
| 25–34 | 2749 | 2175 |
| 35–44 | 2629 | 2103 |
| 45–54 | 2581 | 2103 |
| 55–64 | 2581 | 2079 |
| 65–74 | 2342 | 1912 |
| 75 and over | 2294 | 1840 |

(Source: adapted from https://propranolols.blogspot.com/2020/02/daily-nutritional-requirements-for_9.html)

(i) Describe the patterns in the energy requirements for males and for females.

**(2)**

(ii) Which of these gives a reason for the difference between the energy required by 25-year-old males and the energy required by 25-year-old females?

**(1)**

- ☒ **A** females use more muscle mass than males
- ☒ **B** females eat less than males
- ☒ **C** males eat less than females
- ☒ **D** males use more muscle mass than females

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## 8. Question 3 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

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(iii) Calculate the percentage difference in the energy requirement of males compared with females in the age range of 35–44 years.

**(3)**

percentage difference = ..... %

**(Total for Question 3 = 15 marks)**

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Working and answer

## 9. Question 3 · part a.ii · 3 marks

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

**3** (a) The table gives the recommended daily dietary intake for different groups of males and females.

| Group | Energy in kJ | Protein in g | Vitamin A in mg | Vitamin C in mg | Iron in mg |
|---|---|---|---|---|---|
| MALES | | | | | |
| 15 to 18 years old | 12 600 | 59 | 0.9 | 50 | 12 |
| 25 to 50 years old | 12 200 | 63 | 0.9 | 60 | 10 |
| FEMALES | | | | | |
| 15 to 18 years old | 9200 | 46 | 0.7 | 50 | 15 |
| 25 to 50 years old | 9200 | 50 | 0.7 | 60 | 15 |

(i) Describe what is meant by a balanced diet.

**(2)**

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

**(3)**

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# 9. Question 3 · part a.ii · 3 marks

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

**(2)**

(iv) State a nutrient, shown in the table, that can help to reduce night blindness.

**(1)**

(v) Red meat and some green vegetables are good sources of iron.

100 g of red meat contains 2.5 mg of iron.

100 g of broccoli contains 0.70 mg of iron.

A male in the age range 15 to 18 years eats 250 g of red meat and 200 g of broccoli in one meal.

Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

**(4)**

extra mass = ..... mg

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## 9. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

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# 10. Question 3 · part a.iii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part a.iii only.

**3** (a) The table gives the recommended daily dietary intake for different groups of males and females.

| Group | Energy in kJ | Protein in g | Vitamin A in mg | Vitamin C in mg | Iron in mg |
|---|---|---|---|---|---|
| MALES | | | | | |
| 15 to 18 years old | 12 600 | 59 | 0.9 | 50 | 12 |
| 25 to 50 years old | 12 200 | 63 | 0.9 | 60 | 10 |
| FEMALES | | | | | |
| 15 to 18 years old | 9200 | 46 | 0.7 | 50 | 15 |
| 25 to 50 years old | 9200 | 50 | 0.7 | 60 | 15 |

(i) Describe what is meant by a balanced diet.

**(2)**

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

**(3)**

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## 10. Question 3 · part a.iii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

**(2)**

.....

.....

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.....

(iv) State a nutrient, shown in the table, that can help to reduce night blindness.

**(1)**

.....

(v) Red meat and some green vegetables are good sources of iron.

100 g of red meat contains 2.5 mg of iron.

100 g of broccoli contains 0.70 mg of iron.

A male in the age range 15 to 18 years eats 250 g of red meat and 200 g of broccoli in one meal.

Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

**(4)**

extra mass = ..... mg

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## 10. Question 3 · part a.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# Practice workbook · Explain how tooth decay happens and justify ways to prevent it

Sugar, bacteria and acid in plaque, why molars decay most, and brushing, fluoride, mouthwash, diet and dental visits, including decay data between regions or years.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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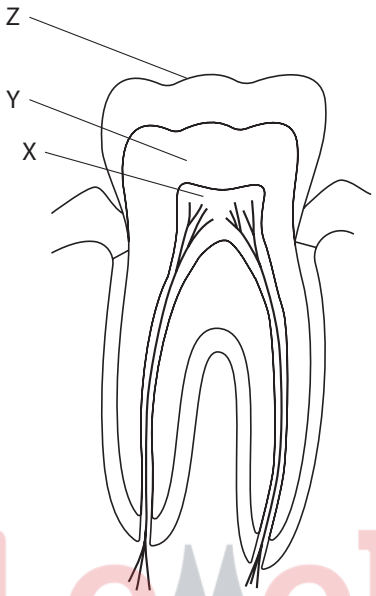
Type ID: HBI-TOOTH-DECAY-AND-CARE

# 1. Question 7 · part b · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b only.

**7** The diagram shows a section through a tooth.




(a) (i) Name the parts of the tooth labelled X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Explain where decay in the tooth is most likely to occur.

**(3)**

.....

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**20**

# 1. Question 7 · part b · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b only.

DO NOT WRITE IN THIS AREA

(b) Explain the role of bacteria in the process of tooth decay.

**(4)**

DO NOT WRITE IN THIS AREA

(c) Babies are usually born without teeth. The teeth develop many months after birth.

Explain why the teeth do not develop until many months after birth.

**(3)**

DO NOT WRITE IN THIS AREA

(d) Dentists recommend using a mouthwash morning and night to reduce dental decay.

One type of mouthwash contains alcohol and sodium fluoride.

Explain how this mouthwash could help to reduce dental decay.

**(2)**

**(Total for Question 7 = 15 marks)**

## 1. Question 7 · part b · 4 marks · Working and answer

Working and answer

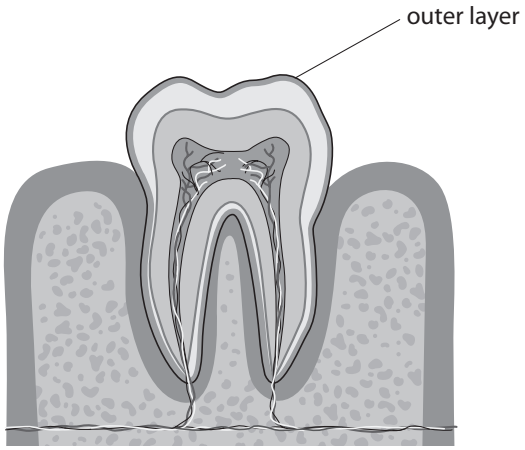
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# 2. Question 1 · part a.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

**1** (a) The diagram shows a section through one type of tooth.




(i) State the name of this type of tooth.

**(1)**

.....

(ii) Which substance forms the outer layer of the tooth?

**(1)**

- ☒ **A** bone
- ☒ **B** dentine
- ☒ **C** enamel
- ☒ **D** pulp

(iii) Explain why this type of tooth is more likely to decay than other types of teeth.

**(3)**

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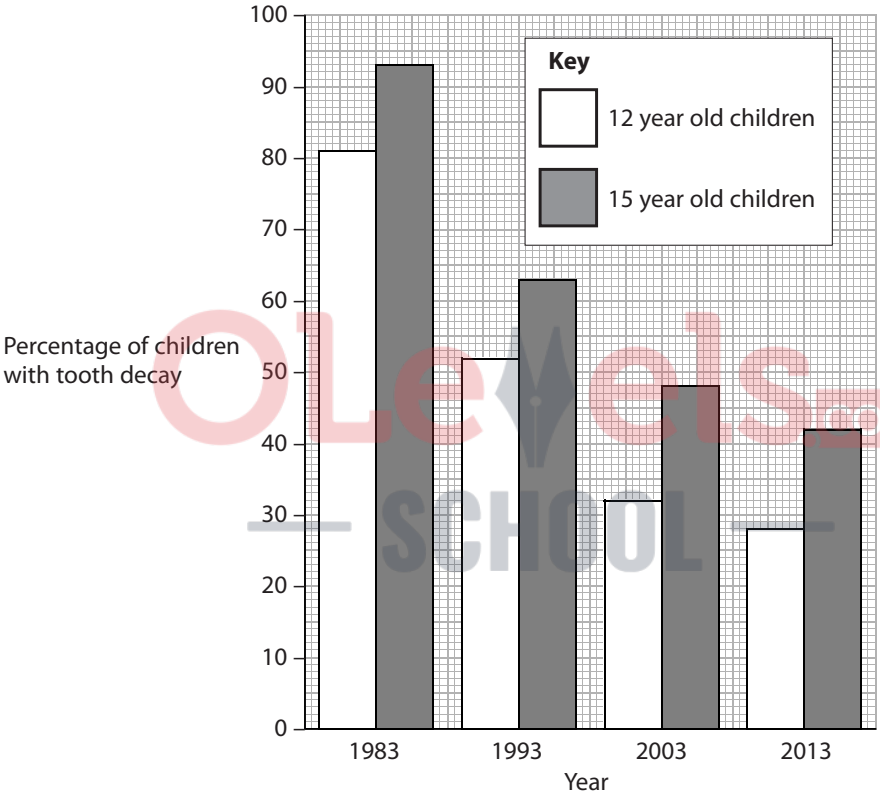
# 2. Question 1 · part a.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

(b) The bar chart shows the percentage of children with tooth decay.

It gives this information for 12 and 15 year old children over a period of 30 years.




(i) Give two conclusions that can be made from the data in the bar chart.

**(2)**

1 .....

.....

2 .....

.....

(ii) Suggest a reason for the change in the percentage of children with tooth decay over the 30 year period.

**(1)**

.....

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**(Total for Question 1 = 8 marks)**

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## 2. Question 1 · part a.iii · 3 marks · Working and answer

Working and answer

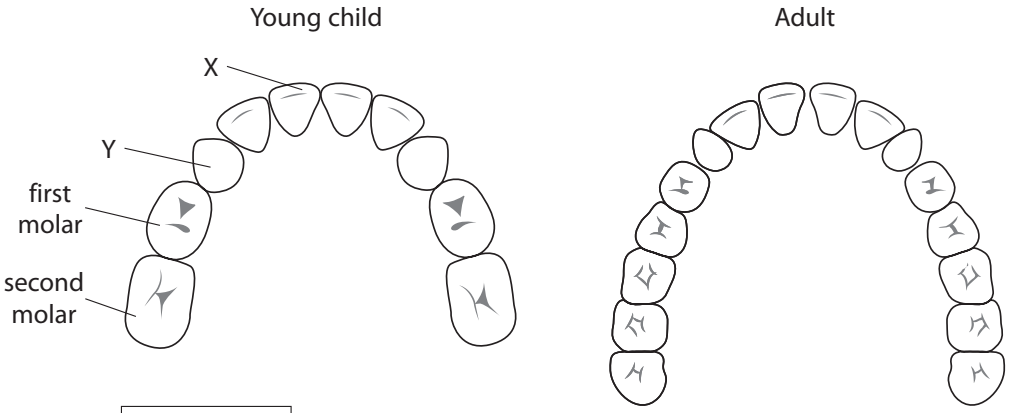
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## 3. Question 1 · part b.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part b.ii only.

**1** The diagram shows the upper teeth of a young child and the upper teeth of an adult.




not to scale

(a) (i) Name tooth X and tooth Y. **(2)**

tooth X .....

tooth Y .....

(ii) State the function of the molar teeth. **(1)**

.....

.....

(iii) State two differences between the teeth of a young child and the teeth of an adult. **(2)**

1 .....

.....

2 .....

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## 3. Question 1 · part b.ii · 2 marks

January 2020 Paper 2

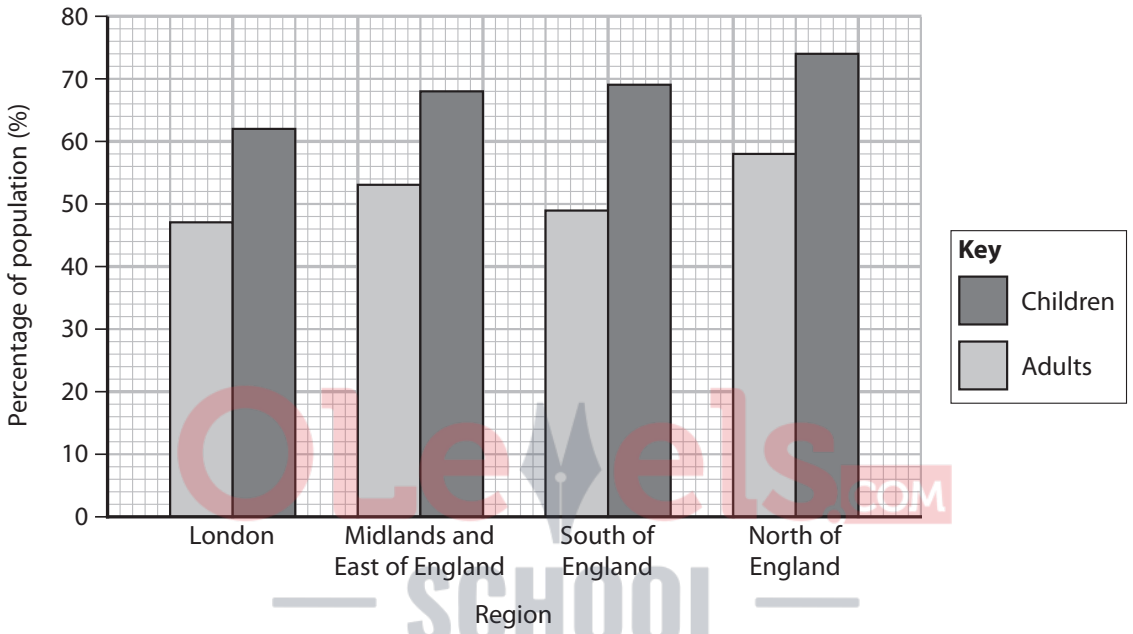
Further practice · Attempt part b.ii only.

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(b) The graph shows the percentage of adults and children in different regions of England who visited a dentist in a two-year period.




Data source/s: Health and Social Care Information Centre NHS Dental Activity Statistics: December 2013 and GP Patient Survey Dental Results 2013/14

(i) Explain which region is likely to have more adults and children with tooth decay.

**(2)**

.....

.....

.....

.....

(ii) Give two reasons why molars are more likely to be damaged by tooth decay than other teeth.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 1 = 9 marks)**

## 3. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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# 4. Question 4 · part c.i · 4 marks

November 2021 Paper 2

Further practice · Attempt part c.i only.

**4** The table shows the results of a survey about the number of teeth per student in a class of young students.

| Number of teeth | Number of students |
|---|---|
| 23 | 1 |
| 24 | 2 |
| 25 | 3 |
| 26 | 4 |
| 27 | 7 |
| 28 | 13 |

(a) (i) Draw a bar chart to show the results of the survey.

**(4)**



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# 4. Question 4 · part c.i · 4 marks

November 2021 Paper 2

Further practice · Attempt part c.i only.

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(ii) A full set of teeth for the students is 28

Calculate the percentage of the students in the class who have a full set of teeth.

**(3)**

percentage = ..... %

(b) A full set of teeth for an adult is 32

Suggest why the full set of teeth for these students is only 28

**(2)**

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# 4. Question 4 · part c.i · 4 marks

November 2021 Paper 2

Further practice · Attempt part c.i only.

(c) (i) Explain how reducing the amount of sugar in foods helps prevent tooth decay.

**(4)**

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(ii) State two other methods that could reduce tooth decay.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 4 = 15 marks)**

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## 4. Question 4 · part c.i · 4 marks · Working and answer

Working and answer

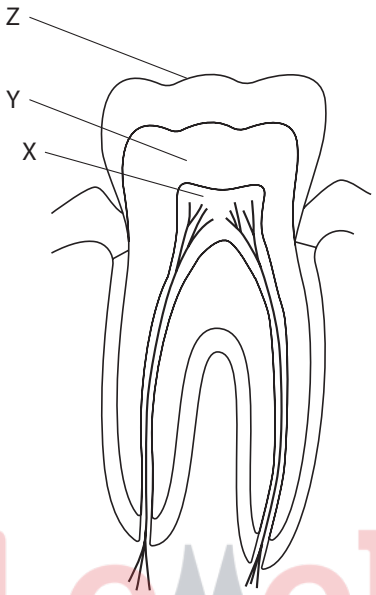
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# 5. Question 7 · part a.ii · 3 marks

June 2023 Paper 2

Further practice · Attempt part a.ii only.

**7** The diagram shows a section through a tooth.




(a) (i) Name the parts of the tooth labelled X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Explain where decay in the tooth is most likely to occur.

**(3)**

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**20**

## 5. Question 7 · part a.ii · 3 marks

June 2023 Paper 2

Further practice · Attempt part a.ii only.

(b) Explain the role of bacteria in the process of tooth decay.

**(4)**

(c) Babies are usually born without teeth. The teeth develop many months after birth.

Explain why the teeth do not develop until many months after birth.

**(3)**

(d) Dentists recommend using a mouthwash morning and night to reduce dental decay.

One type of mouthwash contains alcohol and sodium fluoride.

Explain how this mouthwash could help to reduce dental decay.

**(2)**

**(Total for Question 7 = 15 marks)**

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## 5. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

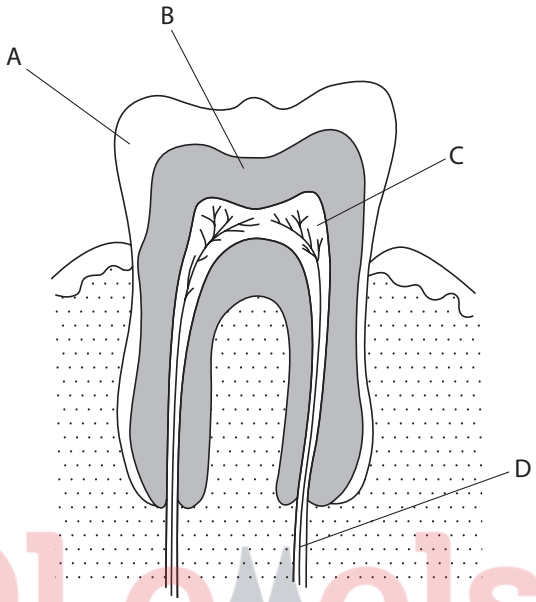
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# 6. Question 3 · part a.iii · 3 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

**3** (a) The diagram shows a tooth.




(i) Name the type of tooth.

**(1)**

.....

(ii) Identify the parts of the tooth labelled A, B, C and D.

**(4)**

A .....

B .....

C .....

D .....

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# 6. Question 3 · part a.iii · 3 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

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(iii) Explain why this type of tooth is more likely to have tooth decay than other types of teeth.

**(3)**

(b) Explain why the minerals obtained from a balanced diet are important in developing and maintaining healthy teeth.

**(4)**

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(c) State three other ways to keep teeth healthy.

**(3)**

1 .....

2 .....

3 .....

**(Total for Question 3 = 15 marks)**

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## 6. Question 3 · part a.iii · 3 marks · Working and answer

Working and answer

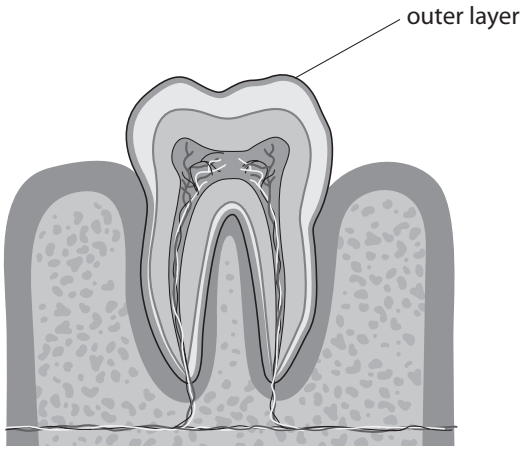
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# 7. Question 1 · part b.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.ii only.

**1** (a) The diagram shows a section through one type of tooth.




(i) State the name of this type of tooth.

**(1)**

(ii) Which substance forms the outer layer of the tooth?

**(1)**

- ☒ **A** bone
- ☒ **B** dentine
- ☒ **C** enamel
- ☒ **D** pulp

(iii) Explain why this type of tooth is more likely to decay than other types of teeth.

**(3)**

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# 7. Question 1 · part b.ii · 1 mark

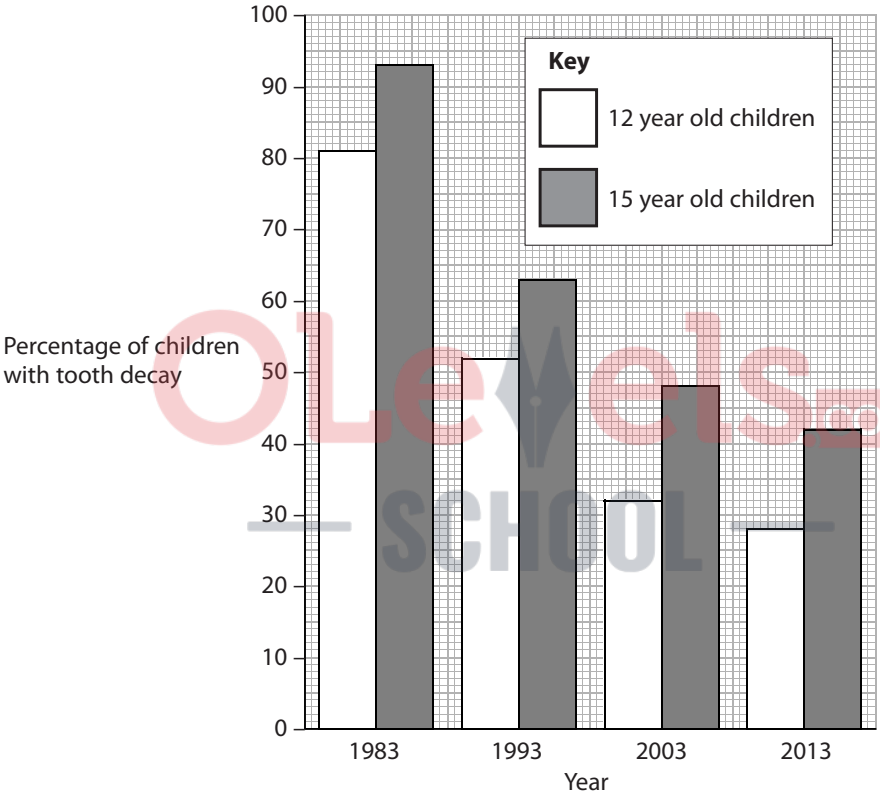
June 2019 Paper 2

Further practice · Attempt part b.ii only.

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(b) The bar chart shows the percentage of children with tooth decay.

It gives this information for 12 and 15 year old children over a period of 30 years.




(i) Give two conclusions that can be made from the data in the bar chart.

**(2)**

1 .....

.....

2 .....

.....

(ii) Suggest a reason for the change in the percentage of children with tooth decay over the 30 year period.

**(1)**

.....

.....

**(Total for Question 1 = 8 marks)**

## 7. Question 1 · part b.ii · 1 mark · Working and answer

Working and answer

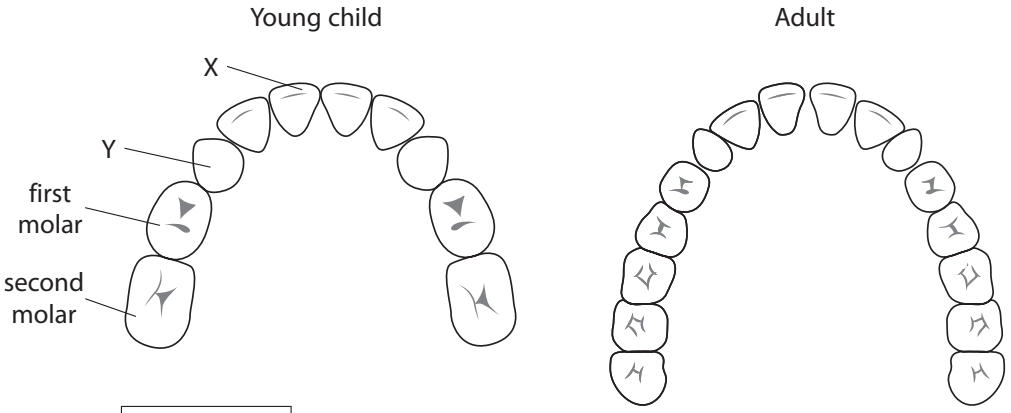
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# 8. Question 1 · part b.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part b.i only.

**1** The diagram shows the upper teeth of a young child and the upper teeth of an adult.




not to scale

(a) (i) Name tooth X and tooth Y. **(2)**

tooth X .....

tooth Y .....

(ii) State the function of the molar teeth. **(1)**

.....

.....

(iii) State two differences between the teeth of a young child and the teeth of an adult. **(2)**

1 .....

.....

2 .....

.....

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# 8. Question 1 · part b.i · 2 marks

January 2020 Paper 2

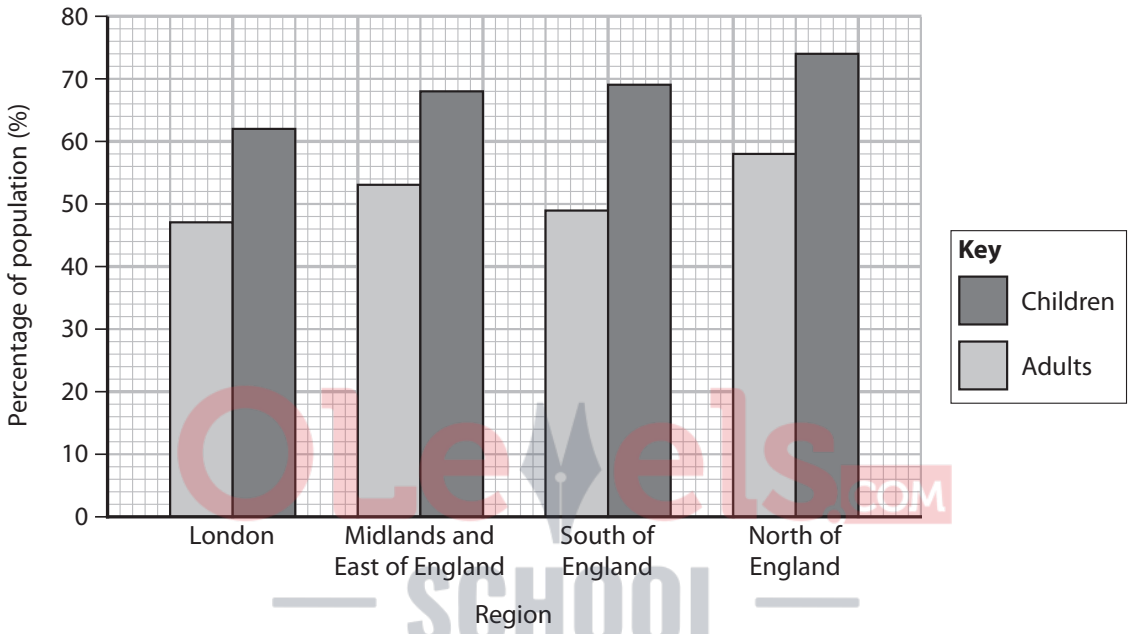
Further practice · Attempt part b.i only.

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(b) The graph shows the percentage of adults and children in different regions of England who visited a dentist in a two-year period.




Data source/s: Health and Social Care Information Centre NHS Dental Activity Statistics: December 2013 and GP Patient Survey Dental Results 2013/14

(i) Explain which region is likely to have more adults and children with tooth decay.

**(2)**

.....

.....

.....

.....

(ii) Give two reasons why molars are more likely to be damaged by tooth decay than other teeth.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 1 = 9 marks)**

## 8. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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# 9. Question 4 · part c.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part c.ii only.

**4** The table shows the results of a survey about the number of teeth per student in a class of young students.

| Number of teeth | Number of students |
|---|---|
| 23 | 1 |
| 24 | 2 |
| 25 | 3 |
| 26 | 4 |
| 27 | 7 |
| 28 | 13 |

(a) (i) Draw a bar chart to show the results of the survey.

**(4)**



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# 9. Question 4 · part c.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part c.ii only.

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(ii) A full set of teeth for the students is 28

Calculate the percentage of the students in the class who have a full set of teeth.

**(3)**

percentage = ..... %

(b) A full set of teeth for an adult is 32

Suggest why the full set of teeth for these students is only 28

**(2)**

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.....

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## 9. Question 4 · part c.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part c.ii only.

(c) (i) Explain how reducing the amount of sugar in foods helps prevent tooth decay.

**(4)**

(ii) State two other methods that could reduce tooth decay.

**(2)**

1 .....

2 .....

**(Total for Question 4 = 15 marks)**

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## 9. Question 4 · part c.ii · 2 marks · Working and answer

Working and answer

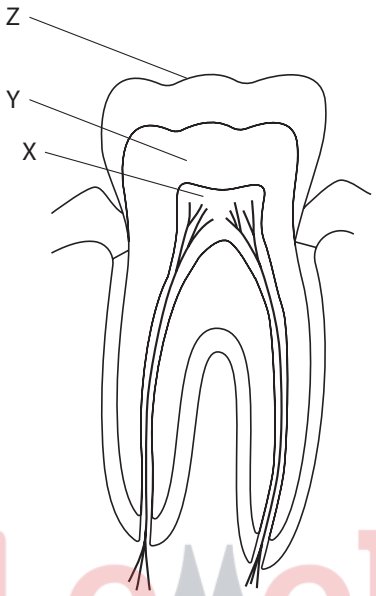
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## 10. Question 7 · part d · 2 marks

June 2023 Paper 2

Further practice · Attempt part d only.

**7** The diagram shows a section through a tooth.




(a) (i) Name the parts of the tooth labelled X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Explain where decay in the tooth is most likely to occur.

**(3)**

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## 10. Question 7 · part d · 2 marks

June 2023 Paper 2

Further practice · Attempt part d only.

(b) Explain the role of bacteria in the process of tooth decay.

**(4)**

(c) Babies are usually born without teeth. The teeth develop many months after birth.

Explain why the teeth do not develop until many months after birth.

**(3)**

(d) Dentists recommend using a mouthwash morning and night to reduce dental decay.

One type of mouthwash contains alcohol and sodium fluoride.

Explain how this mouthwash could help to reduce dental decay.

**(2)**

**(Total for Question 7 = 15 marks)**

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## 10. Question 7 · part d · 2 marks · Working and answer

Working and answer

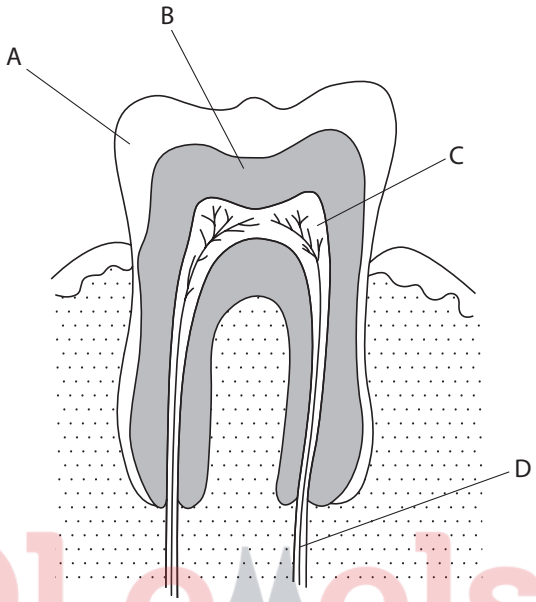
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## 11. Question 3 · part b · 4 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

**3** (a) The diagram shows a tooth.




(i) Name the type of tooth.

**(1)**

.....

(ii) Identify the parts of the tooth labelled A, B, C and D.

**(4)**

A .....

B .....

C .....

D .....

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# 11. Question 3 · part b · 4 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

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(iii) Explain why this type of tooth is more likely to have tooth decay than other types of teeth.

**(3)**

(b) Explain why the minerals obtained from a balanced diet are important in developing and maintaining healthy teeth.

**(4)**

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(c) State three other ways to keep teeth healthy.

**(3)**

1 .....

2 .....

3 .....

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**(Total for Question 3 = 15 marks)**

## 11. Question 3 · part b · 4 marks · Working and answer

Working and answer

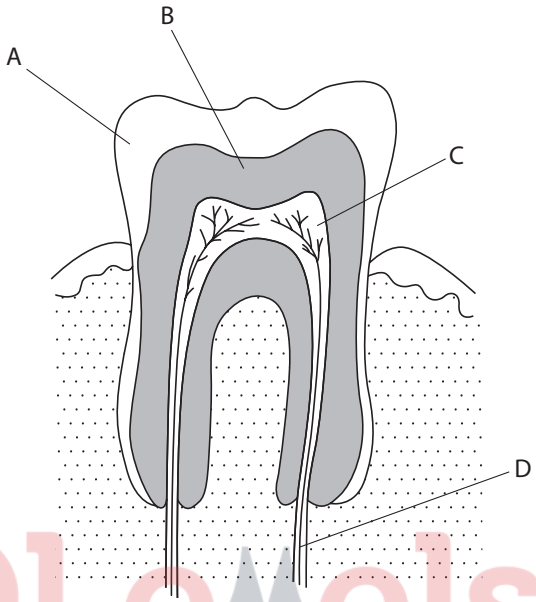
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## 12. Question 3 · part c · 3 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

**3** (a) The diagram shows a tooth.




(i) Name the type of tooth.

**(1)**

.....

(ii) Identify the parts of the tooth labelled A, B, C and D.

**(4)**

A .....

B .....

C .....

D .....

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P 7 4 6 2 8 A 0 6 2 0

## 12. Question 3 · part c · 3 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

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(iii) Explain why this type of tooth is more likely to have tooth decay than other types of teeth.

**(3)**

(b) Explain why the minerals obtained from a balanced diet are important in developing and maintaining healthy teeth.

**(4)**

OLevels.com SCHOOL

(c) State three other ways to keep teeth healthy.

**(3)**

1 .....

2 .....

3 .....

**(Total for Question 3 = 15 marks)**

## 12. Question 3 · part c · 3 marks · Working and answer

Working and answer

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Practice workbook · Explain an enzyme practical result, including levelling off as substrate runs out

Observations such as falling pH, a positive sugar test, a clearing film or a plateau in product, explained by digestion and by substrate becoming limiting.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ENZYME-RESULT-EXPLAIN

# 1. Question 7 · part b.ii · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

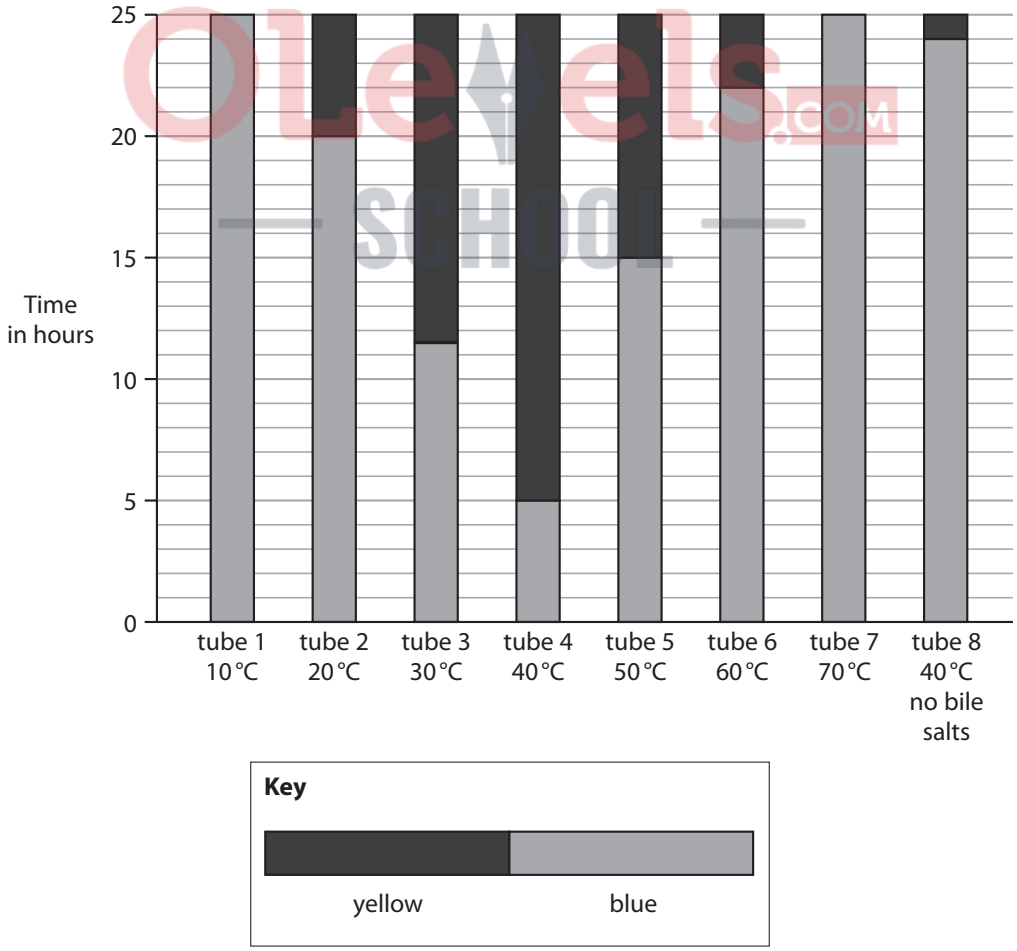
**7** A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.




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# 1. Question 7 · part b.ii · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

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(a) (i) State three factors that should be kept constant in the student's investigation.

**(3)**

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

**(1)**

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

**(3)**

.....

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(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

**(2)**

time = ..... hours

(ii) Explain why the pH decreases during the 25 hours.

**(3)**

.....

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.....

.....

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.....

# 1. Question 7 · part b.ii · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

(c) Explain the result for tube 8.

**(3)**

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**(Total for Question 7 = 15 marks)**

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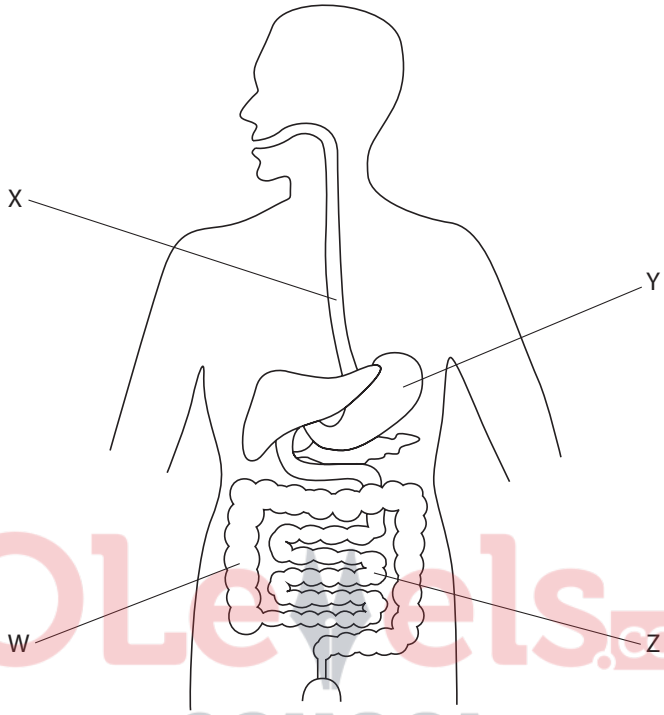
Working and answer

# 2. Question 2 · part b.iii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.iii only.

**2** (a) The diagram shows the human digestive system.




(i) Name the structures labelled W, X, Y and Z.

**(4)**

W .....

X .....

Y .....

Z .....

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P 7 0 9 4 9 A 0 6 2 4

## 2. Question 2 · part b.iii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.iii only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

| | | | |
|---|---|---|---|
| relax | coordination | contract | liquid |
| peristalsis | muscles | enzymes | bolus |

Use words from the box to complete the passage.

**(4)**

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

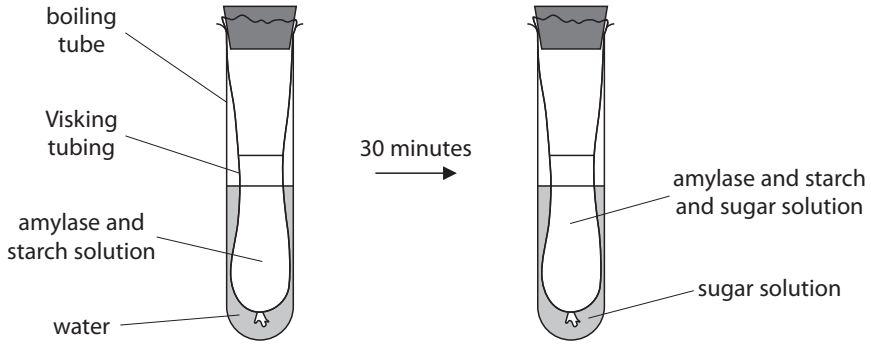
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# 2. Question 2 · part b.iii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.iii only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.




(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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## 2. Question 2 · part b.iii · 2 marks · Working and answer

Working and answer

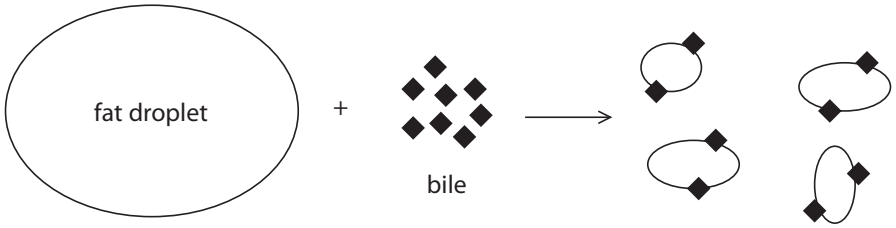
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# 3. Question 3 · part b.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.ii only.

**3** (a) The diagram represents the effect of bile on a large fat droplet.




(i) Where is bile produced?

**(1)**

- ☒ **A** gall bladder
- ☒ **B** liver
- ☒ **C** small intestine
- ☒ **D** stomach

(ii) Describe the role of bile in the digestion of fat.

**(3)**

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# 3. Question 3 · part b.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.ii only.

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(b) Three food tests are used on a solution of food.

The table shows the reagent used and the colour of the food solution after each test.

| Reagent used | Colour of the food solution after the test |
|---|---|
| iodine solution | blue/black |
| biuret solution | purple |
| Benedict's solution | blue |

(i) Describe what the food tests show about the dietary components in the food.

**(3)**

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(ii) Saliva is added to a new sample of the food solution and the mixture is kept at 37 °C for 10 minutes.

The food tests are then repeated and the solution turns brick red when tested with Benedict's reagent.

Explain the effect of saliva on the food solution.

**(3)**

# 3. Question 3 · part b.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.ii only.

(c) Food is digested as it moves along the alimentary canal.

This process is helped by a diet containing fibre.

Explain how food is moved along the alimentary canal and how fibre helps this process.

**(5)**

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**(Total for Question 3 = 15 marks)**

## 3. Question 3 · part b.ii · 3 marks · Working and answer

Working and answer

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# 4. Question 2 · part b · 3 marks

November 2024 Paper 1

Further practice · Attempt part b only.

**2** A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

| Time in minutes | pH of solution |
|---|---|
| 1 | 6.8 |
| 2 | 6.4 |
| 3 | 6.0 |
| 4 | 5.8 |
| 5 | 5.6 |
| 6 | 5.2 |
| 7 | 5.0 |
| 8 | 4.8 |
| 9 | 4.6 |
| 10 | 4.6 |

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# 4. Question 2 · part b · 3 marks

November 2024 Paper 1

Further practice · Attempt part b only.

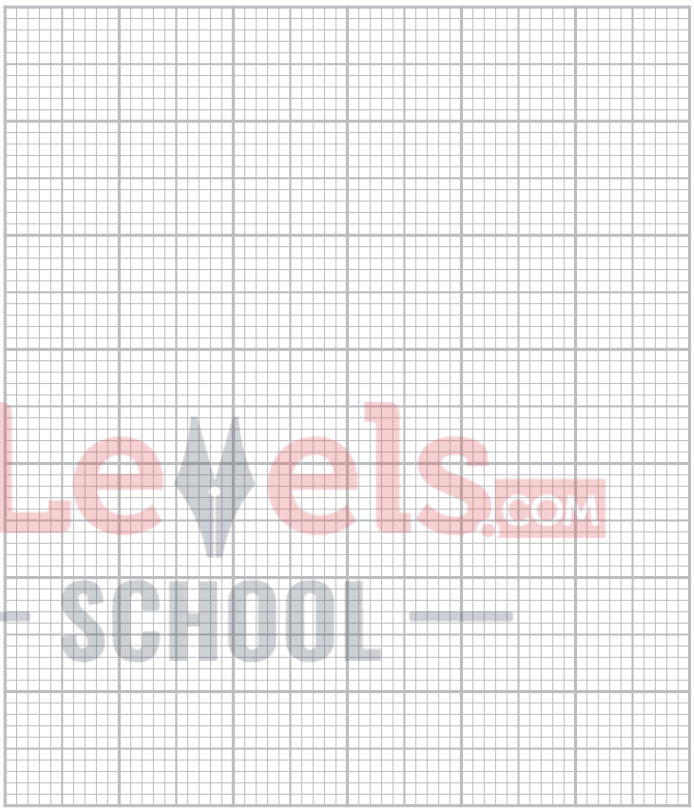
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(a) (i) Plot a graph of the student's results.

**(3)**



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(ii) Draw a line of best fit.

**(1)**

(iii) Give the trend shown by the graph.

**(1)**

.....

.....

## 4. Question 2 · part b · 3 marks

November 2024 Paper 1

Further practice · Attempt part b only.

(b) Explain the difference in the pH at 1 minute and the pH at 10 minutes.

**(3)**

(c) Name the independent variable in this investigation.

**(1)**

(d) Give two variables that the student should control in this investigation.

**(2)**

1 .....

2 .....

(e) State how the student could improve the reliability of their results.

**(1)**

**(Total for Question 2 = 12 marks)**

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## 4. Question 2 · part b · 3 marks · Working and answer

Working and answer

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# 5. Question 4 · part b.i · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.i only.

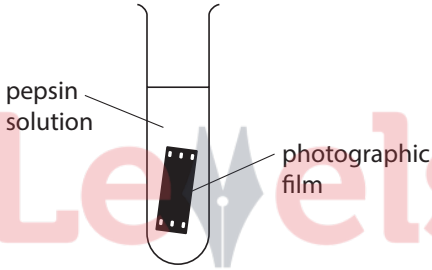
**4** Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin




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The table shows the student's results.

| Percentage pepsin concentration (%) | Time taken for film to become clear in seconds |
|---|---|
| 1 | 112 |
| 2 | 102 |
| 3 | 93 |
| 4 | 84 |
| 5 | 84 |
| 6 | 84 |

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## 5. Question 4 · part b.i · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.i only.

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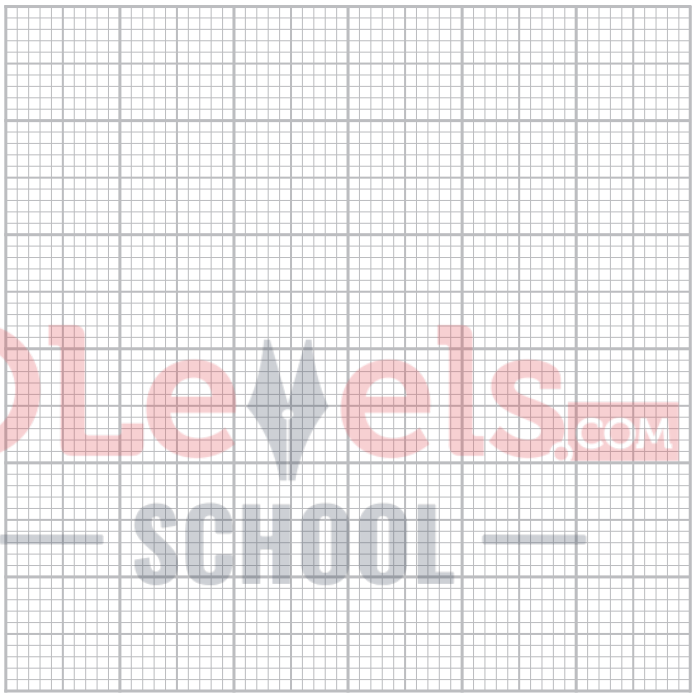
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(a) (i) Plot a graph of the results on the grid.

**(4)**

(ii) Draw a curve of best fit.

**(1)**



(b) (i) Explain why the photographic film becomes clear.

**(3)**

.....

.....

.....

.....

## 5. Question 4 · part b.i · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.i only.

(ii) Explain which would be the best pH to carry out this investigation.

**(2)**

(iii) State three variables that should be controlled during this investigation other than pH.

**(3)**

1 .....

2 .....

3 .....

(c) Suggest why the time taken for the film to become clear is the same for 4%, 5% and 6% concentrations of pepsin.

**(2)**

**(Total for Question 4 = 15 marks)**

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## 5. Question 4 · part b.i · 3 marks · Working and answer

Working and answer

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# 6. Question 7 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

**7** Cheese is processed from milk. During this process, an enzyme is used to convert the lactose in milk into lactic acid.

A student uses this method to investigate the process.

- dissolve lactose in water to produce a solution
- put the lactose solution into three different test tubes
- keep the first test tube at 2 °C, the second test tube at 25 °C and the third test tube at 55 °C
- add enzyme solution to the lactose solution in each test tube
- measure the amount of lactic acid in the solution every 10 hours for a total of 60 hours

The table shows the student's results.

| Time after adding enzyme in hours | Amount of lactic acid in the solution in arbitrary units | | |
|---|---|---|---|
| | 2 °C | 25 °C | 55 °C |
| 0 | 0.1 | 0.1 | 0.1 |
| 10 | 0.1 | 0.2 | 0.2 |
| 20 | 0.1 | 0.3 | 0.2 |
| 30 | 0.2 | 0.8 | 0.4 |
| 40 | 0.3 | 1.8 | 0.7 |
| 50 | 0.4 | 2.2 | 1.0 |
| 60 | 0.6 | 2.3 | 1.3 |

(a) Describe what additional information another student would need to repeat this investigation.

**(4)**

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**20**

## 6. Question 7 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

(b) (i) Estimate the optimum temperature for this enzyme.

**(1)**

(ii) Explain why it is not possible to give an accurate value for the optimum temperature using the student's method.

**(2)**

(c) The student continued the investigation for a further 60 hours.

Explain the results that would be obtained.

**(3)**

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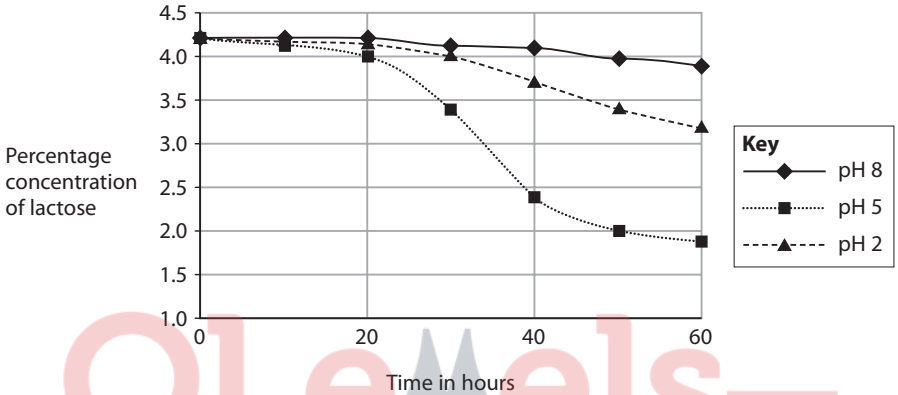
## 6. Question 7 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

(d) The student uses the same enzyme to investigate the effect of pH on the conversion of the lactose into lactic acid.

The student's results are shown in the graph.




The student wants to determine the most effective conditions for the conversion of lactose into lactic acid in cheese manufacture.

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## 6. Question 7 · part c · 3 marks · Working and answer

Working and answer

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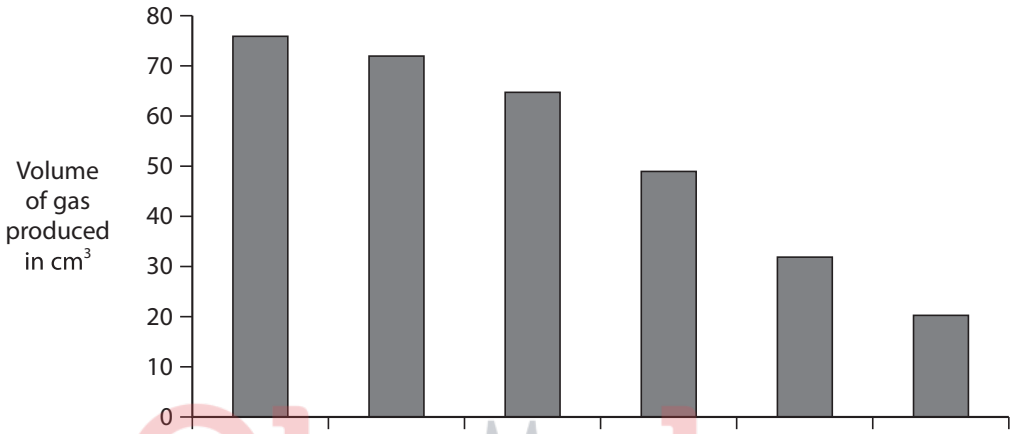
# 7. Question 3 · part a.iv · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iv only.

**3** A student investigates a reaction. The reaction is catalysed by an enzyme and produces a gas. The student collects the gas for 10 minutes and measures the volume of the gas. The student repeats this every 10 minutes for one hour.

The bar chart shows the student's results.




(a) (i) Add a suitable labelled scale to the *x*-axis.

**(2)**

(ii) Estimate the total volume of gas produced after 30 minutes.

**(2)**

volume of gas = ..... $cm^3$

(iii) Describe the trend shown by the results.

**(3)**

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# 7. Question 3 · part a.iv · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iv only.

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(iv) Explain the student's results obtained in the investigation.

**(3)**

(b) Explain the effect on the results if the student had added a competitive inhibitor at the start of the investigation.

**(2)**

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**(Total for Question 3 = 12 marks)**

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## 7. Question 3 · part a.iv · 3 marks · Working and answer

Working and answer

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# 8. Question 4 · part c · 2 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

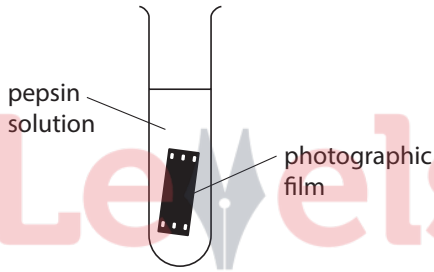
**4** Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin




The table shows the student's results.

| Percentage pepsin concentration (%) | Time taken for film to become clear in seconds |
|---|---|
| 1 | 112 |
| 2 | 102 |
| 3 | 93 |
| 4 | 84 |
| 5 | 84 |
| 6 | 84 |

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## 8. Question 4 · part c · 2 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

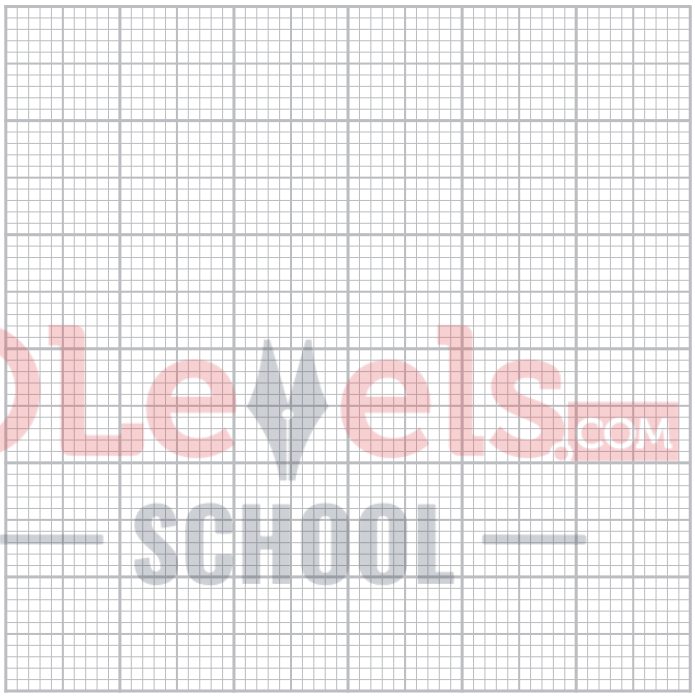
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(a) (i) Plot a graph of the results on the grid.

**(4)**

(ii) Draw a curve of best fit.

**(1)**



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(b) (i) Explain why the photographic film becomes clear.

**(3)**

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.....

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P 7 4 6 2 8 A 0 9 2 0

## 8. Question 4 · part c · 2 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

(ii) Explain which would be the best pH to carry out this investigation.

**(2)**

(iii) State three variables that should be controlled during this investigation other than pH.

**(3)**

1 .....

2 .....

3 .....

(c) Suggest why the time taken for the film to become clear is the same for 4%, 5% and 6% concentrations of pepsin.

**(2)**

**(Total for Question 4 = 15 marks)**

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## 8. Question 4 · part c · 2 marks · Working and answer

Working and answer

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# Practice workbook · Explain how villi aid absorption and the effect of damage

Villus and microvillus structure for absorption, and consequences when villi are flattened or nutrients are taken by parasites.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-VILLUS-ABSORPTION

# 1. Question 5 · part c.ii · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part c.ii only.

**5** Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

**(3)**

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

**(1)**

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

**(1)**

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# 1. Question 5 · part c.ii · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part c.ii only.

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(iii) Draw a diagram to show how the student could use the equipment to investigate the effect of glucose concentration on the rate of diffusion.

You should add these labels to your diagram.

- Visking tubing
- boiling tube
- sugar solution
- water

**(3)**

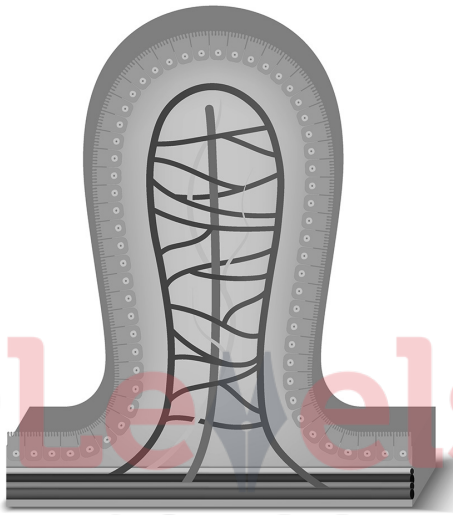
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# 1. Question 5 · part c.ii · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part c.ii only.

(c) The diagram shows a structure found in the small intestine. This structure increases the rate of diffusion of molecules such as glucose from the small intestine into the blood stream.



(Source: © Dee-sign / Shutterstock)

(i) Name this structure.

**(1)**

.....

(ii) Explain how these structures increase the rate of diffusion of glucose from the small intestine into the blood stream.

**(3)**

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**(Total for Question 5 = 12 marks)**

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## 1. Question 5 · part c.ii · 3 marks · Working and answer

Working and answer

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## 2. Question 4 · part c.iii · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.iii only.

**4** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

**(1)**

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

**(2)**

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**8**

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## 2. Question 4 · part c.iii · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.iii only.

(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

**(2)**

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

**(1)**

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

**(1)**

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

**(2)**

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

**(1)**

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## 2. Question 4 · part c.iii · 2 marks · Working and answer

Working and answer

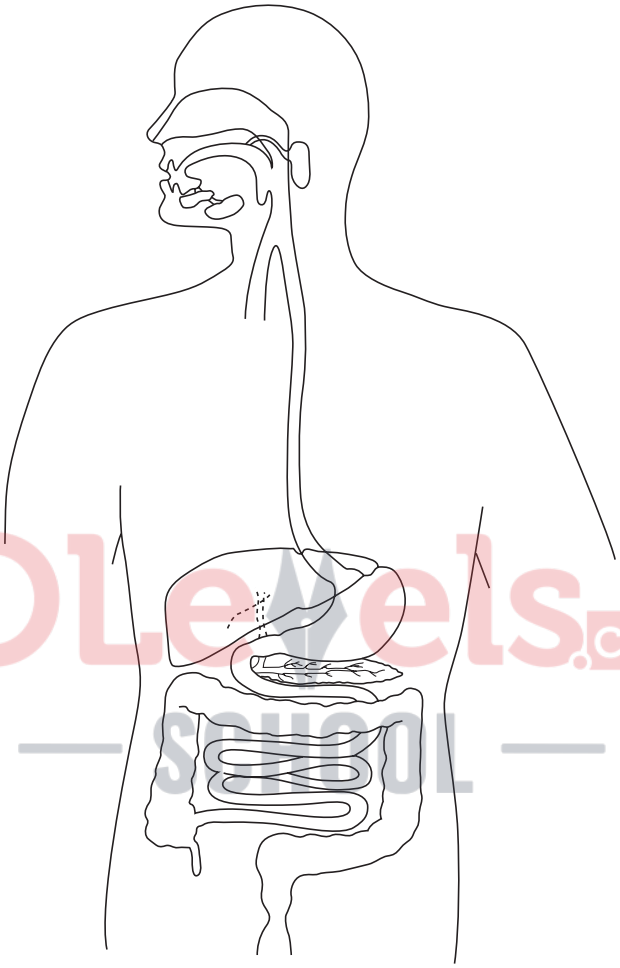
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# 3. Question 4 · part b · 4 marks

November 2021 Paper 1

Further practice · Attempt part b only.

**4** The diagram shows the alimentary canal.



(a) Identify these areas of the alimentary canal using label lines and the correct letters.

(i) Label with an X the area where starch digestion begins.

**(1)**

(ii) Label with a Y the area that has a pH of 2

**(1)**

(iii) Label with a Z the area where both lipid and protein digesting enzymes are produced.

**(1)**

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## 3. Question 4 · part b · 4 marks

November 2021 Paper 1

Further practice · Attempt part b only.

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(b) Some people have the condition known as coeliac disease. This results in the villi being flattened.

Explain why some people with untreated coeliac disease will eventually develop osteoporosis.

**(4)**

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(c) Another condition of the alimentary canal is colon cancer. This is often treated by removal of the large intestine.

Explain why people who have this treatment often suffer from dehydration.

**(2)**

**(Total for Question 4 = 9 marks)**

## 3. Question 4 · part b · 4 marks · Working and answer

Working and answer

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# 4. Question 6 · part c.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

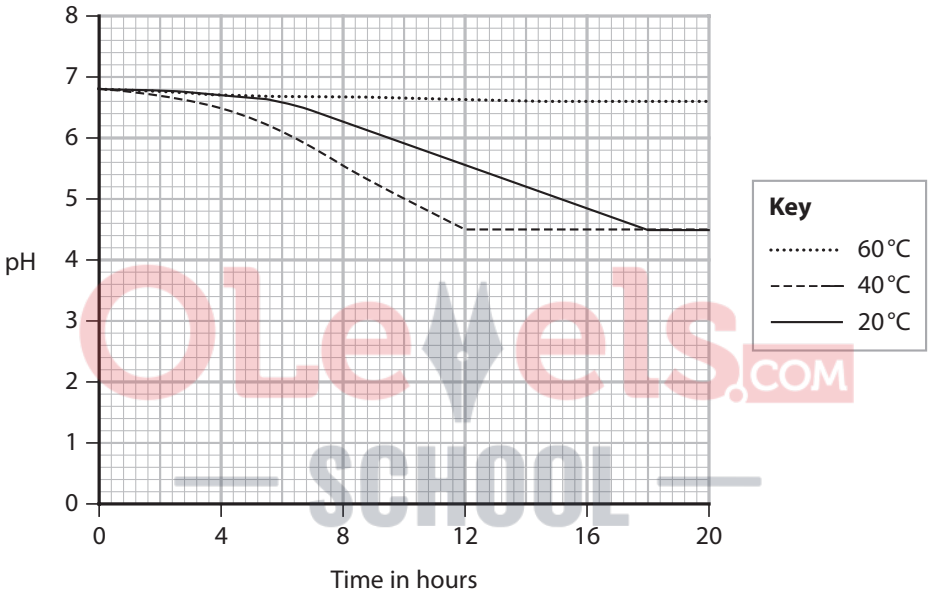
**6** Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.




(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

**(2)**

time = ..... hours

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# 4. Question 6 · part c.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

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(ii) Give two variables that need to be controlled in this investigation.

**(2)**

1 .....

.....

2 .....

.....

(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

**(2)**

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(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

**(2)**

boiling

.....

.....

cooling

.....

.....



P 7 2 2 7 1 A 0 1 7 2 4

# 4. Question 6 · part c.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

(b) When yoghurt reaches a pH of 4.4 the bacteria stop fermenting lactose.

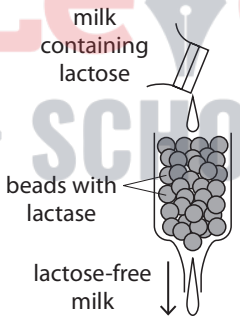
Explain why bacteria stop fermenting the lactose at a pH of around 4.4

**(2)**

(c) People who are lactose intolerant can drink lactose-free milk.

This can be produced using alginate beads containing lactase.

The diagram shows the equipment needed.




(i) Describe the advantages of using alginate beads containing lactase rather than a solution of lactase.

**(2)**

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# 4. Question 6 · part c.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

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(ii) The enzyme lactase breaks down lactose into simpler sugars.

Explain how the structure of the small intestine allows the absorption of these sugars.

**(3)**

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**(Total for Question 6 = 15 marks)**

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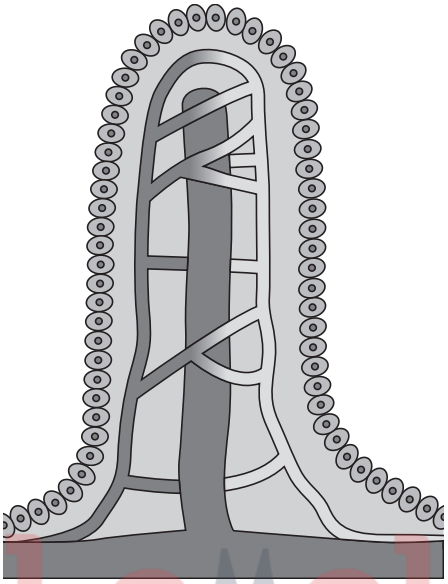
Working and answer

# 5. Question 7 · part a · 3 marks

June 2023 Paper 2R

Further practice · Attempt part a only.

**7** The diagram shows a villus in the small intestine of the digestive system.



(a) Explain how the villus is adapted to its function in the digestive system.

**(3)**

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## 5. Question 7 · part a · 3 marks

June 2023 Paper 2R

Further practice · Attempt part a only.

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(b) (i) Coeliac disease is an autoimmune disease.

The immune system of people with coeliac disease attacks the body's own tissue and damages the lining of the small intestine.

Describe how the immune system of people with coeliac disease damages the lining of the small intestine.

**(2)**

(ii) Explain how coeliac disease affects the absorption of nutrients in the small intestine.

**(3)**

**(Total for Question 7 = 8 marks)**

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## 5. Question 7 · part a · 3 marks · Working and answer

Working and answer

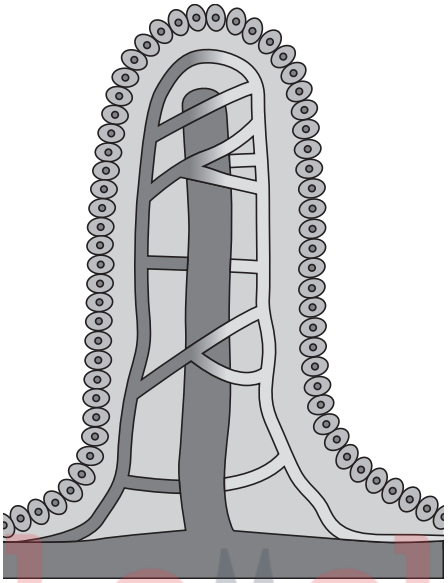
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# 6. Question 7 · part b.ii · 3 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

**7** The diagram shows a villus in the small intestine of the digestive system.



(a) Explain how the villus is adapted to its function in the digestive system.

**(3)**

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# 6. Question 7 · part b.ii · 3 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

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(b) (i) Coeliac disease is an autoimmune disease.

The immune system of people with coeliac disease attacks the body's own tissue and damages the lining of the small intestine.

Describe how the immune system of people with coeliac disease damages the lining of the small intestine.

**(2)**

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(ii) Explain how coeliac disease affects the absorption of nutrients in the small intestine.

**(3)**

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**(Total for Question 7 = 8 marks)**

OLevels.COM — SCHOOL —

## 6. Question 7 · part b.ii · 3 marks · Working and answer

Working and answer

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# 7. Question 5 · part c.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part c.i only.

**5** Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

**(3)**

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

**(1)**

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

**(1)**

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P 7 4 6 3 2 A 0 1 4 2 4

## 7. Question 5 · part c.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part c.i only.

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(iii) Draw a diagram to show how the student could use the equipment to investigate the effect of glucose concentration on the rate of diffusion.

You should add these labels to your diagram.

- Visking tubing
- boiling tube
- sugar solution
- water

**(3)**

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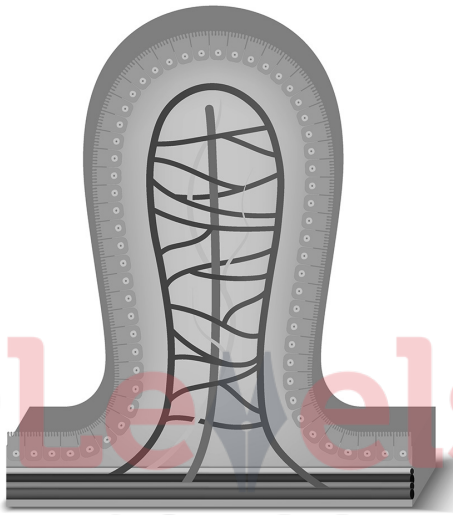
P 7 4 6 3 2 A 0 1 5 2 4

# 7. Question 5 · part c.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part c.i only.

(c) The diagram shows a structure found in the small intestine. This structure increases the rate of diffusion of molecules such as glucose from the small intestine into the blood stream.



(Source: © Dee-sign / Shutterstock)

(i) Name this structure.

**(1)**

.....

(ii) Explain how these structures increase the rate of diffusion of glucose from the small intestine into the blood stream.

**(3)**

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**(Total for Question 5 = 12 marks)**

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## 7. Question 5 · part c.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

# Practice workbook · Explain where bile comes from and how it aids fat digestion

Bile production in the liver, storage in the gall bladder, emulsification of fat and neutralising stomach acid, including liver damage and bile-free tubes.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

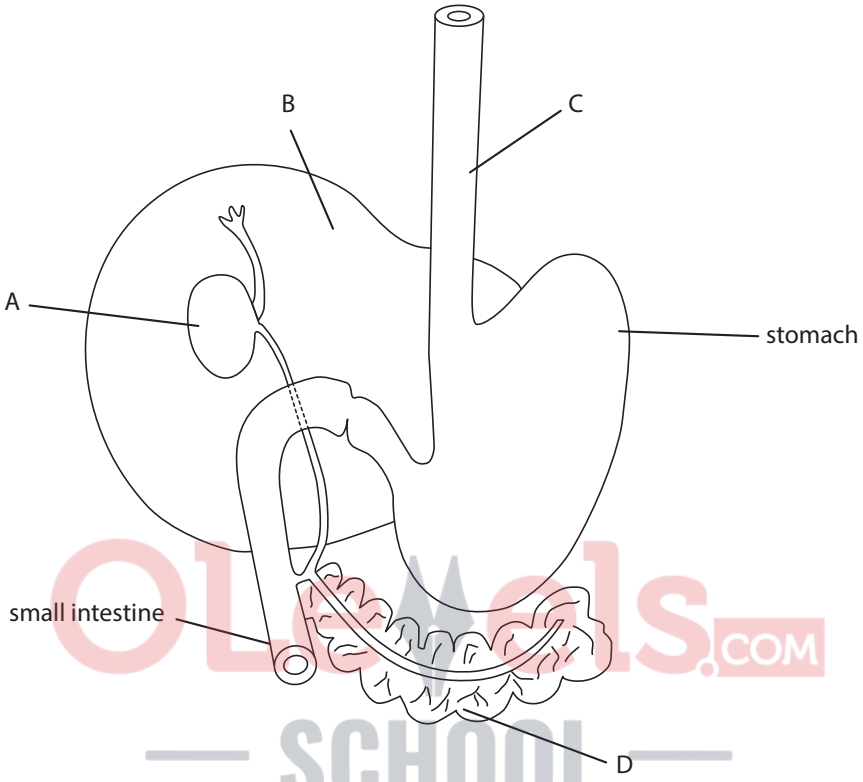
Type ID: HBI-BILE-EMULSIFY-NEUTRALISE

# 1. Question 2 · part a.iii · 3 marks

June 2025 Paper 2

Worked-example retry · Attempt part a.iii only.

**2** The diagram shows structures involved in digestion.




(a) (i) Identify the structures labelled A, B, C and D.

**(4)**

A .....

B .....

C .....

D .....

(ii) Name the fluid stored in structure A.

**(1)**

.....

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P 7 8 7 6 9 A 0 4 2 0

# 1. Question 2 · part a.iii · 3 marks

June 2025 Paper 2

Worked-example retry · Attempt part a.iii only.

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(iii) Explain the function of the fluid stored in structure A in digestion.

**(3)**

(b) (i) Describe the role of structure D in digestion.

**(4)**

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(ii) Name two substances secreted by structure D that are not involved in digestion.

**(2)**

1 .....

2 .....

**(Total for Question 2 = 14 marks)**

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## 1. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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## 2. Question 7 · part c · 5 marks

June 2019 Paper 2

Further practice · Attempt part c only.

**7** The volume of alcohol in alcoholic drinks can be measured in units.

One unit is equivalent to 10 $cm^3$ of pure alcohol.

The diagram shows a bottle of wine with 14% alcohol content and a glass containing 250 $cm^3$ of this wine.




250 $cm^3$ glass of Ruby Red wine

(a) Calculate the number of units of alcohol in the glass of wine.

**(2)**

number of units = .....

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## 2. Question 7 · part c · 5 marks

June 2019 Paper 2

Further practice · Attempt part c only.

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(b) It is illegal for a person to drive a motor vehicle if they have more than the legal limit of alcohol in their bloodstream.

It takes one hour for the body to process one unit of alcohol so that there is no alcohol left in the bloodstream.

(i) A person drinks three units of alcohol. This puts her at the legal limit.

Calculate the time it would take before a person can legally drive after drinking four glasses of Ruby Red wine.

**(2)**

time = ..... hours

(ii) Suggest why it is illegal for people to drive a motor vehicle with more than the legal limit of alcohol in their bloodstream.

**(2)**

.....

.....

.....

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## 2. Question 7 · part c · 5 marks

June 2019 Paper 2

Further practice · Attempt part c only.

(c) Drinking a lot of alcohol over a long period of time can cause a condition called cirrhosis.

This condition prevents the cells of the liver from functioning normally.

Explain the effect of cirrhosis on the digestion of food molecules.

**(5)**

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**(Total for Question 7 = 11 marks)**

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## 2. Question 7 · part c · 5 marks · Working and answer

Working and answer

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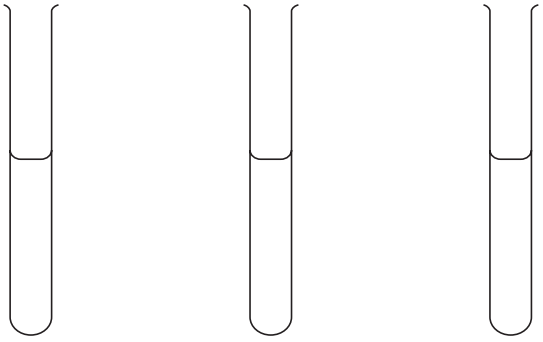
# 3. Question 2 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

**2** A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.



**A** **B** **C**

Table 1 lists the contents of each test tube.

| Test tube A | Test tube B | Test tube C |
|---|---|---|
| 5 $cm^3$ of milk | 5 $cm^3$ of milk | 5 $cm^3$ of milk |
| 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate |
| 6 drops of phenolphthalein | 6 drops of phenolphthalein | 6 drops of phenolphthalein |
| 1 $cm^3$ enzyme | 1 $cm^3$ enzyme | 1 $cm^3$ boiled enzyme |
| distilled water | bile salts | bile salts |

**Table 1**

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# 3. Question 2 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

| | Colour of contents | | |
|---|---|---|---|
| | **Tube A** | **Tube B** | **Tube C** |
| at start | pink | pink | pink |
| after 5 minutes | pink | colourless | pink |
| after 10 minutes | pink | colourless | pink |
| after 15 minutes | colourless | colourless | pink |

**Table 2**

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation? **(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids? **(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced? **(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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## 3. Question 2 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

(b) Explain the purpose of tube C.

**(2)**

(c) Suggest one reason for the addition of the sodium hydrogencarbonate.

**(2)**

(d) Describe the effect of bile salts on lipid digestion in this investigation.

**(3)**

**(Total for Question 2 = 10 marks)**

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## 3. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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# 4. Question 2 · part d · 3 marks

November 2020 Paper 2

Further practice · Attempt part d only.

**2** A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.

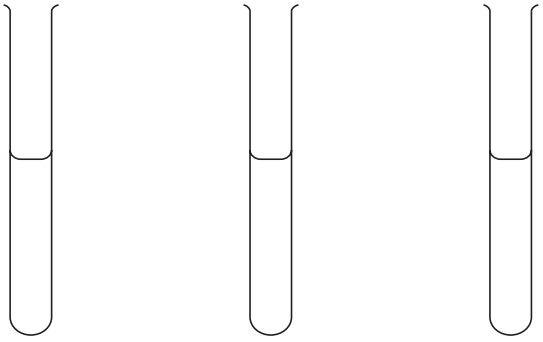



Table 1 lists the contents of each test tube.

| Test tube A | Test tube B | Test tube C |
|---|---|---|
| 5 $cm^3$ of milk | 5 $cm^3$ of milk | 5 $cm^3$ of milk |
| 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate |
| 6 drops of phenolphthalein | 6 drops of phenolphthalein | 6 drops of phenolphthalein |
| 1 $cm^3$ enzyme | 1 $cm^3$ enzyme | 1 $cm^3$ boiled enzyme |
| distilled water | bile salts | bile salts |

**Table 1**

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## 4. Question 2 · part d · 3 marks

November 2020 Paper 2

Further practice · Attempt part d only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

| | Colour of contents | | |
|---|---|---|---|
| | **Tube A** | **Tube B** | **Tube C** |
| at start | pink | pink | pink |
| after 5 minutes | pink | colourless | pink |
| after 10 minutes | pink | colourless | pink |
| after 15 minutes | colourless | colourless | pink |

**Table 2**

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation?

**(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids?

**(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced?

**(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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# 4. Question 2 · part d · 3 marks

November 2020 Paper 2

Further practice · Attempt part d only.

(b) Explain the purpose of tube C.

**(2)**

(c) Suggest one reason for the addition of the sodium hydrogencarbonate.

**(2)**

(d) Describe the effect of bile salts on lipid digestion in this investigation.

**(3)**

**(Total for Question 2 = 10 marks)**

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## 4. Question 2 · part d · 3 marks · Working and answer

Working and answer

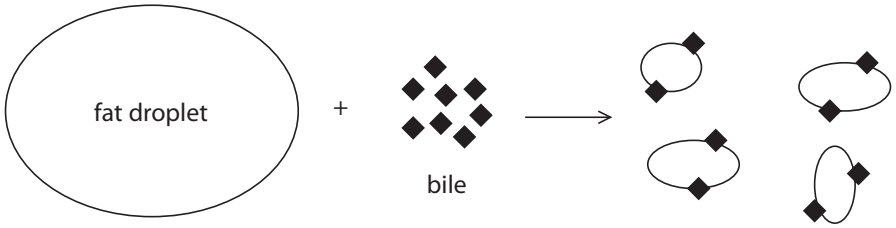
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## 5. Question 3 · part a.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.i only.

**3** (a) The diagram represents the effect of bile on a large fat droplet.




(i) Where is bile produced?

**(1)**

- ☒ **A** gall bladder
- ☒ **B** liver
- ☒ **C** small intestine
- ☒ **D** stomach

(ii) Describe the role of bile in the digestion of fat.

**(3)**

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## 5. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

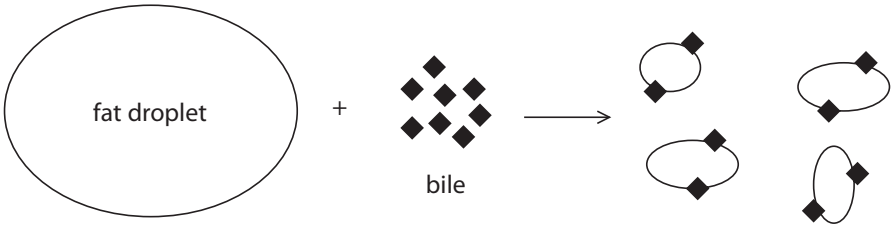
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## 6. Question 3 · part a.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.ii only.

**3** (a) The diagram represents the effect of bile on a large fat droplet.




(i) Where is bile produced?

**(1)**

- ☒ **A** gall bladder
- ☒ **B** liver
- ☒ **C** small intestine
- ☒ **D** stomach

(ii) Describe the role of bile in the digestion of fat.

**(3)**

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# 6. Question 3 · part a.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.ii only.

(b) Three food tests are used on a solution of food.

The table shows the reagent used and the colour of the food solution after each test.

| Reagent used | Colour of the food solution after the test |
|---|---|
| iodine solution | blue/black |
| biuret solution | purple |
| Benedict's solution | blue |

(i) Describe what the food tests show about the dietary components in the food.

**(3)**

(ii) Saliva is added to a new sample of the food solution and the mixture is kept at 37 °C for 10 minutes.

The food tests are then repeated and the solution turns brick red when tested with Benedict's reagent.

Explain the effect of saliva on the food solution.

**(3)**

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## 6. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

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# 7. Question 7 · part c · 3 marks

November 2023 Paper 2

Further practice · Attempt part c only.

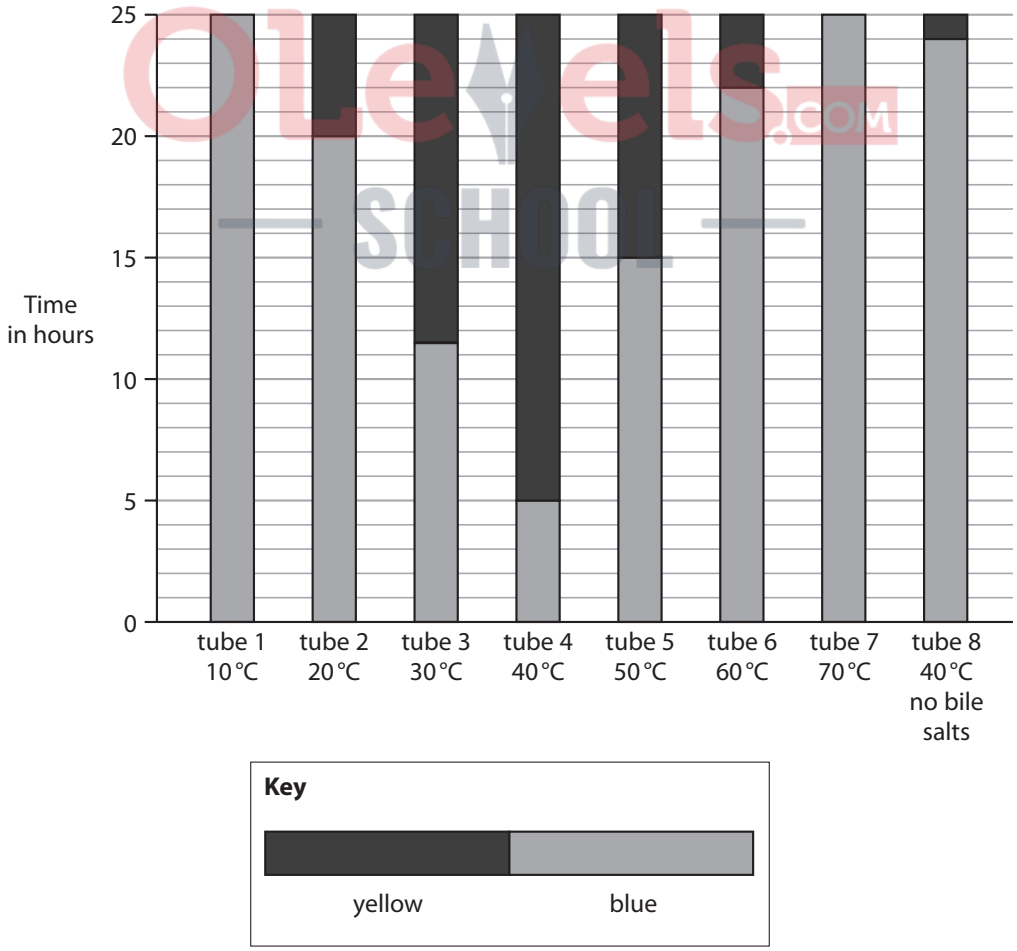
**7** A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.




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# 7. Question 7 · part c · 3 marks

November 2023 Paper 2

Further practice · Attempt part c only.

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(a) (i) State three factors that should be kept constant in the student's investigation.

**(3)**

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

**(1)**

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

**(3)**

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(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

**(2)**

time = ..... hours

(ii) Explain why the pH decreases during the 25 hours.

**(3)**

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# 7. Question 7 · part c · 3 marks

November 2023 Paper 2

Further practice · Attempt part c only.

(c) Explain the result for tube 8.

**(3)**

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**(Total for Question 7 = 15 marks)**

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Working and answer

# 8. Question 3 · part b · 2 marks

June 2024 Paper 2

Further practice · Attempt part b only.

**3** Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

**(2)**

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## 8. Question 3 · part b · 2 marks

June 2024 Paper 2

Further practice · Attempt part b only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

**(2)**

(c) State how cataracts affect vision (Lines 4 to 5).

**(1)**

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

**(1)**

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

**(3)**

## 8. Question 3 · part b · 2 marks · Working and answer

Working and answer

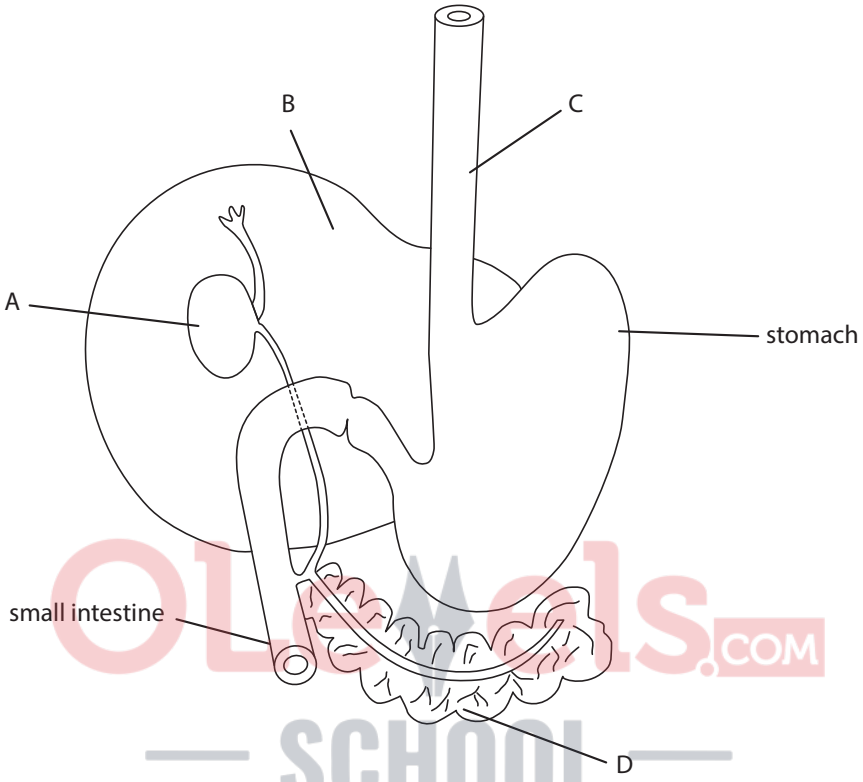
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# 9. Question 2 · part a.ii · 1 mark

June 2025 Paper 2

Further practice · Attempt part a.ii only.

**2** The diagram shows structures involved in digestion.




(a) (i) Identify the structures labelled A, B, C and D.

**(4)**

A .....

B .....

C .....

D .....

(ii) Name the fluid stored in structure A.

**(1)**

.....

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## 9. Question 2 · part a.ii · 1 mark

June 2025 Paper 2

Further practice · Attempt part a.ii only.

(iii) Explain the function of the fluid stored in structure A in digestion.

**(3)**

(b) (i) Describe the role of structure D in digestion.

**(4)**

(ii) Name two substances secreted by structure D that are not involved in digestion.

**(2)**

1 .....

2 .....

**(Total for Question 2 = 14 marks)**

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## 9. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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Practice workbook · Measure food energy by burning and explain why the value is too low

Burning a food sample under a tube or can of water, describing the method, explaining heat losses and how a calorimeter or apparatus changes improve accuracy.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-FOOD-ENERGY-CALORIMETRY

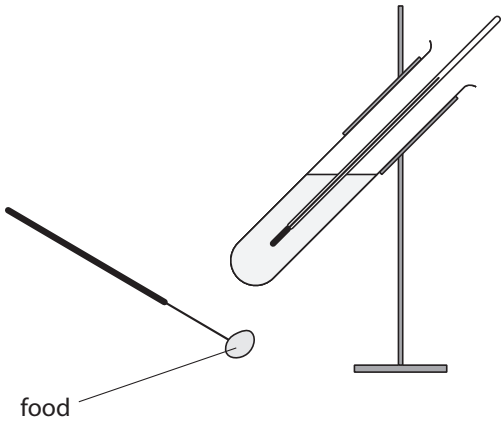
# 1. Question 7 · part a · 6 marks

November 2024 Paper 1

Worked-example retry · Attempt part a only.

**7** In a healthy person, the amount of energy in the daily diet should equal the amount of energy used by the body.

(a) The equipment shown can be used to measure the amount of energy in food.




Describe how this equipment could be used to compare the energy content of different foods.

**(6)**

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# 1. Question 7 · part a · 6 marks

November 2024 Paper 1

Worked-example retry · Attempt part a only.

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(b) Describe a named method of transport that uses energy to move molecules across a cell membrane.

**(2)**

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**(Total for Question 7 = 8 marks)**

Working and answer

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## 2. Question 5 · part b.i · 4 marks

November 2020 Paper 1R

Further practice · Attempt part b.i only.

**5** (a) The type and amount of different nutrients needed for a balanced diet varies depending on several factors.

(i) Pregnant women are often advised by their doctor to take extra iron in their diet.

Which two foods are the best sources of iron?

**(1)**

- ☒ **A** fruit and green vegetables
- ☒ **B** dairy products and red meat
- ☒ **C** green vegetables and red meat
- ☒ **D** fruit and dairy products

(ii) Explain why a woman should have more iron in her diet when she becomes pregnant.

**(4)**

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## 2. Question 5 · part b.i · 4 marks

November 2020 Paper 1R

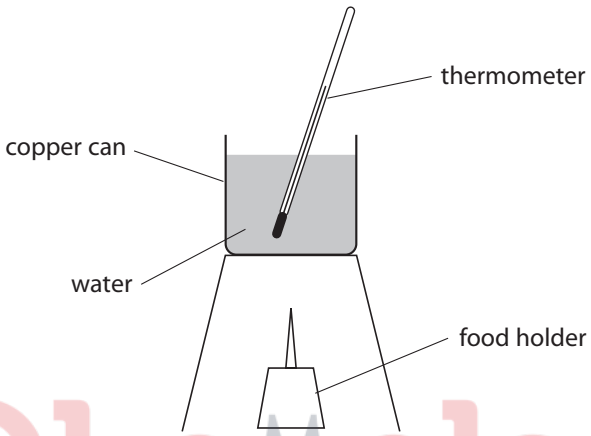
Further practice · Attempt part b.i only.

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(b) The diagram shows some apparatus that can be used to measure the amount of energy contained in different foods.




(i) Explain how the apparatus can be used to obtain data about the energy content of different foods.

**(4)**

(ii) Explain how this apparatus can be changed to improve the accuracy of the data collected for each food.

**(3)**

**(Total for Question 5 = 12 marks)**

## 2. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

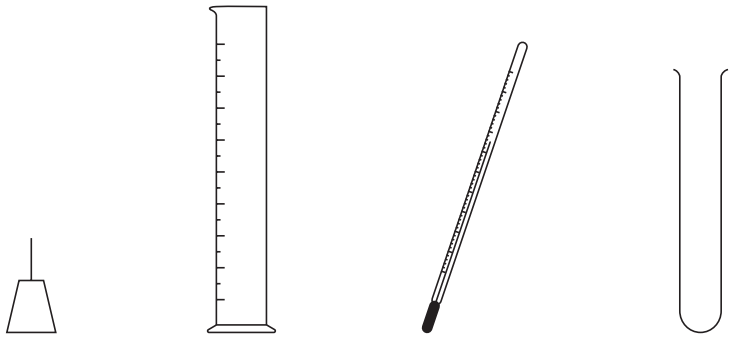
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# 3. Question 3 · part a.i · 5 marks

June 2022 Paper 1

Further practice · Attempt part a.i only.

**3** (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

**(5)**

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# 3. Question 3 · part a.i · 5 marks

June 2022 Paper 1

Further practice · Attempt part a.i only.

(ii) Name two variables that need to be controlled in the investigation.

**(2)**

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(iii) Give two safety precautions needed for this investigation.

**(2)**

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## 3. Question 3 · part a.i · 5 marks · Working and answer

Working and answer

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# 4. Question 5 · part b.ii · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

**5** (a) The type and amount of different nutrients needed for a balanced diet varies depending on several factors.

(i) Pregnant women are often advised by their doctor to take extra iron in their diet.

Which two foods are the best sources of iron?

**(1)**

- ☒ **A** fruit and green vegetables
- ☒ **B** dairy products and red meat
- ☒ **C** green vegetables and red meat
- ☒ **D** fruit and dairy products

(ii) Explain why a woman should have more iron in her diet when she becomes pregnant.

**(4)**

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# 4. Question 5 · part b.ii · 3 marks

November 2020 Paper 1R

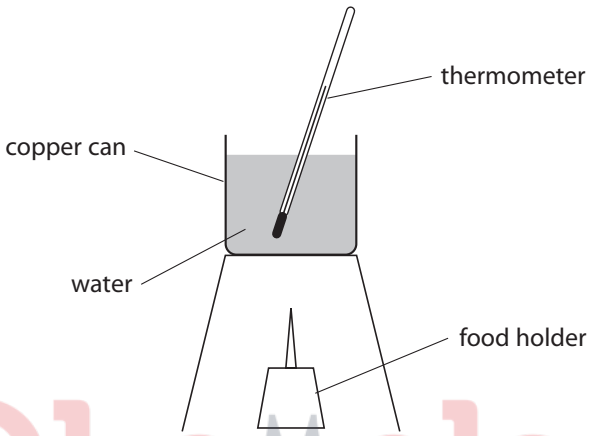
Further practice · Attempt part b.ii only.

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(b) The diagram shows some apparatus that can be used to measure the amount of energy contained in different foods.




(i) Explain how the apparatus can be used to obtain data about the energy content of different foods.

**(4)**

(ii) Explain how this apparatus can be changed to improve the accuracy of the data collected for each food.

**(3)**

**(Total for Question 5 = 12 marks)**

## 4. Question 5 · part b.ii · 3 marks · Working and answer

Working and answer

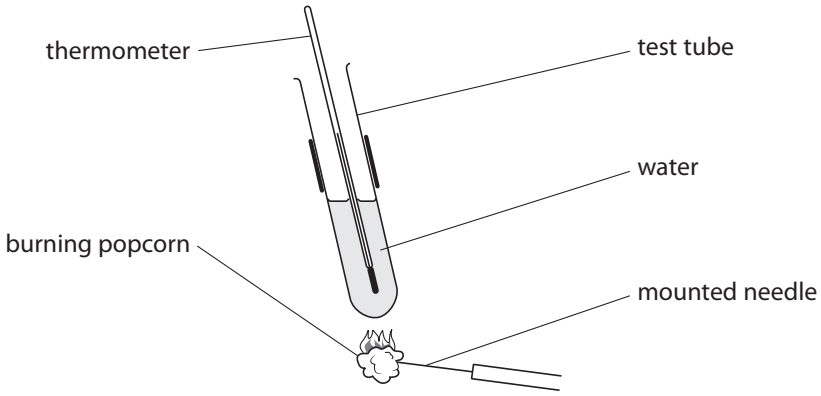
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# 5. Question 2 · part a.i · 1 mark

November 2020 Paper 1

Further practice · Attempt part a.i only.

**2** A student uses this apparatus to investigate the energy contained in popcorn.




This is the student's method.

- place 12 g of water in a test tube
- measure the initial temperature of the water
- place the popcorn on a mounted needle
- light the popcorn using a Bunsen flame
- place the burning popcorn under the test tube
- measure the maximum temperature of the water

(a) (i) State why the test tube should have thin glass walls.

**(1)**

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(ii) Explain one safety precaution that the student should take during the investigation.

**(2)**

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## 5. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

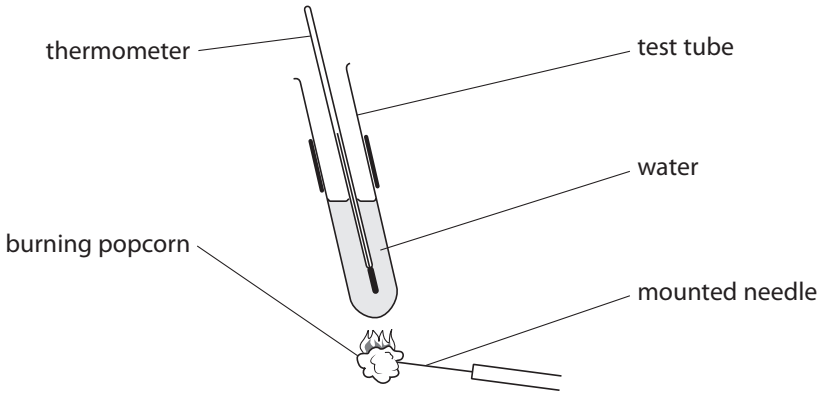
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# 6. Question 2 · part b.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part b.ii only.

**2** A student uses this apparatus to investigate the energy contained in popcorn.




This is the student's method.

- place 12 g of water in a test tube
- measure the initial temperature of the water
- place the popcorn on a mounted needle
- light the popcorn using a Bunsen flame
- place the burning popcorn under the test tube
- measure the maximum temperature of the water

(a) (i) State why the test tube should have thin glass walls.

**(1)**

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(ii) Explain one safety precaution that the student should take during the investigation.

**(2)**

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P 6 2 0 5 1 A 0 4 2 4

## 6. Question 2 · part b.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part b.ii only.

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(b) The student uses this equation to calculate the energy released by the popcorn.

energy released = mass (g) × 4.2 × temperature rise (°C)

The initial temperature of the 12 g of water is 17 °C.

The maximum temperature of the water is 45 °C.

(i) Calculate the energy released when the student burns the popcorn.

Give your answer to 2 significant figures.

**(3)**

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energy released = ..... J

(ii) Explain why the calculated amount of energy released is not the same as the full amount of energy contained in the popcorn.

**(3)**

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## 6. Question 2 · part b.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part b.ii only.

(c) The student wants to compare the energy content of popcorn with the energy content of two other foods.

State four factors that the student should control to make sure the comparison is valid.

**(4)**

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**(Total for Question 2 = 13 marks)**

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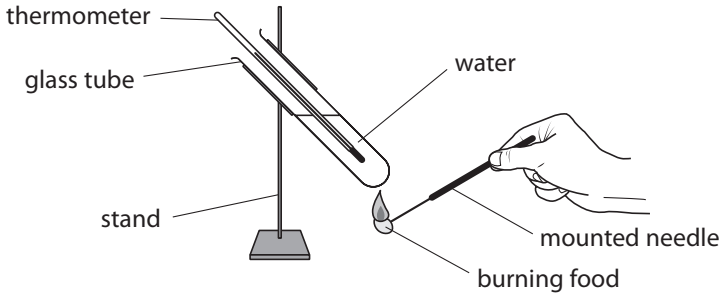
Working and answer

# 7. Question 7 · part a.iii · 5 marks

June 2021 Paper 1

Further practice · Attempt part a.iii only.

**7** A student uses this apparatus to investigate the energy content of different foods.




This is the student's method.

- add water to the glass tube
- measure the temperature of the water
- place a piece of food onto the mounted needle
- set the food alight by placing it into a Bunsen flame
- place the burning food under the glass tube
- measure the temperature of the water when the food stops burning

Repeat the method using different foods.

(a) (i) State three safety precautions that the student should take.

**(3)**

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# 7. Question 7 · part a.iii · 5 marks

June 2021 Paper 1

Further practice · Attempt part a.iii only.

(ii) State three factors that should be controlled during the investigation.

**(3)**

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(iii) Explain why the energy content of a sample of food is likely to be higher than the energy of the same sample measured by this method.

**(5)**

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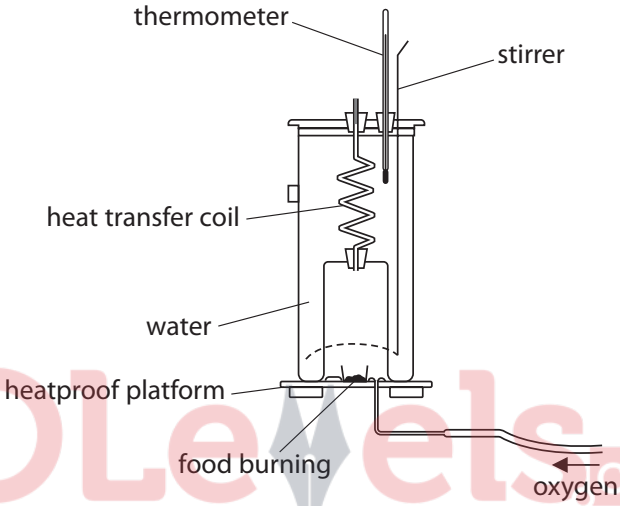
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# 7. Question 7 · part a.iii · 5 marks

June 2021 Paper 1

Further practice · Attempt part a.iii only.

(b) The diagram shows a calorimeter. This is a piece of apparatus that is used to measure the energy content of a sample of food accurately.




Discuss the features of this calorimeter that will give a more accurate measurement of the energy in a sample of food than the method used by the student.

**(4)**

**(Total for Question 7 = 15 marks)**

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## 7. Question 7 · part a.iii · 5 marks · Working and answer

Working and answer

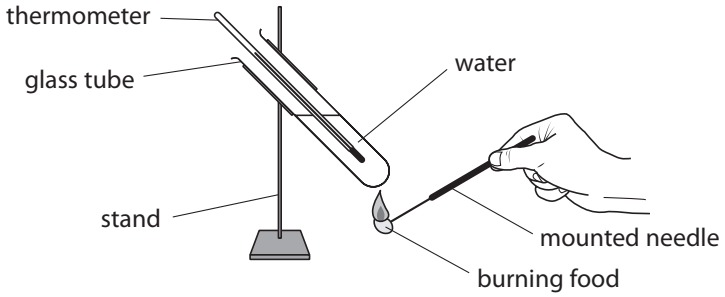
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# 8. Question 7 · part b · 4 marks

June 2021 Paper 1

Further practice · Attempt part b only.

**7** A student uses this apparatus to investigate the energy content of different foods.




This is the student's method.

- add water to the glass tube
- measure the temperature of the water
- place a piece of food onto the mounted needle
- set the food alight by placing it into a Bunsen flame
- place the burning food under the glass tube
- measure the temperature of the water when the food stops burning

Repeat the method using different foods.

(a) (i) State three safety precautions that the student should take.

**(3)**

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## 8. Question 7 · part b · 4 marks

June 2021 Paper 1

Further practice · Attempt part b only.

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(ii) State three factors that should be controlled during the investigation.

**(3)**

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(iii) Explain why the energy content of a sample of food is likely to be higher than the energy of the same sample measured by this method.

**(5)**

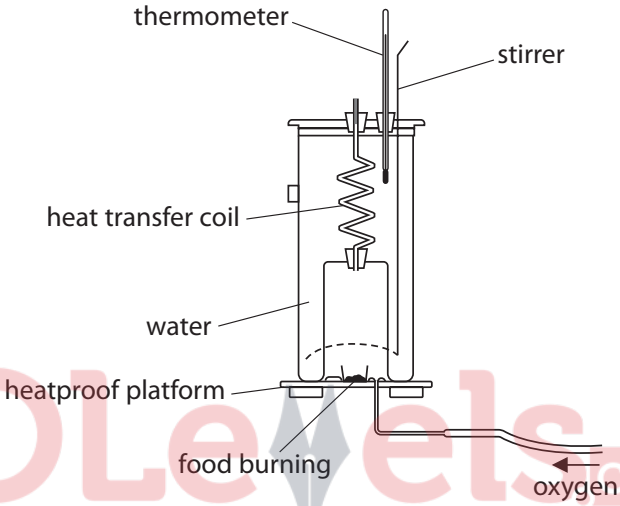
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## 8. Question 7 · part b · 4 marks

June 2021 Paper 1

Further practice · Attempt part b only.

(b) The diagram shows a calorimeter. This is a piece of apparatus that is used to measure the energy content of a sample of food accurately.




Discuss the features of this calorimeter that will give a more accurate measurement of the energy in a sample of food than the method used by the student.

**(4)**

**(Total for Question 7 = 15 marks)**

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## 8. Question 7 · part b · 4 marks · Working and answer

Working and answer

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